14 MW Waste-to-Energy Project by Antony Lara Renewable Energy Pvt. Ltd

Project title	*14 MW Waste-to-Energy Project by Antony Lara Renewable Energy Pvt. Ltd*
Project ID	*4620*
Crediting period	*Starts on 07-October-2023 and ends on 06–October-2030 (both days included).*
Original date of issue	*29-February-2024*
Most recent date of issue	*22-April-2025*
Version	*08*
VCS Standard Version	*4.7*
Prepared by	*Kosher Climate India Private Limited*

CONTENTS

1 PROJECT DETAILS

1.1 Summary Description of the Project

The project activity is 14MW Waste to Energy Power Plant located at Moshi Kachara Depot, Pimpri, Pune, India. The plant utilizes Refuse-Derived Fuel ("RDF") as feedstock derived from the separation and sorting of Municipal Solid Waste ("MSW") excluding the recyclable or potentially hazardous materials from the waste stream. The MSW will be sourced from the Pimpri Chinchwad Municipality area as per concession agreement with Pimpri Chinchwad Municipal Corporation (herein after PCMC).

The PCMC has been providing around 1200 TPD of MSW which is pre-processed in Material Recovery Facility (MRF). After pre-processed/segregation of the MSW, 700 TPD of RDF (pre-processed) such as paper, wood including coconut shell, garden waste, cloth etc with higher combustible properties has been processed by the Antony Lara Renewable Energy Pvt Ltd (ALREPL) for waste to energy plant which comes under the scope of the project activity. The sampling of the RDF/ pre-processed waste has been used for the assessment of physical characterisation of waste. In addition to this, the food waste (organic waste stream) has been utilized for composting for further processing and the inert reject has been disposed in sanitary landfill.

M/s Antony Lara Renewable Energy Pvt.Ltd (ALREPL) includes the construction and operation of RDF based waste to energy plant.

The main purpose of this project activity is to generate clean form of electricity through waste to energy-based power generation. The project activity will treat about 700 ton of RDF per day and generate electricity with the installed capacity of 14 MW. The waste to energy plant will use 1 reciprocating grate firing type boiler of 70TPH steam generation and 1 double bleed cum condensing steam turbine generator of 14MW capacity and is connected to the Indian grid to supply 77,616 MWh/Year of net electricity. Further, PCMC has signed a conversion charges agreement to utilize the electricity generated by the project through the grid.

Scenario existing prior to the implementation of project activity:

This is a greenfield project activity. Prior to the implementation of the project activity, MSW under Pimpri Chinchwad Municipality was mainly subjected to landfill without any methane recovery system for flaring or energy purpose, therefore methane emissions from anaerobic decay of waste would have been released into the atmosphere.

For electricity component, prior to the implementation of the project activity, the electricity supply for Pimpri Chinchwad municipality area was generated mainly from grid connected power plants.

The project activity helps to avoid methane emissions from landfill site in the absence of the project activity and generate renewable electricity, which displaced part of the electricity otherwise supplied by grid connected power plants. The project activity is expected to achieve an annual average emission reduction of **57,719 tCO_2e** over the 7-year crediting period and a total emission reduction of **461,756 tCO_2e** during the first 7-year renewable crediting period.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation/ verification	Crediting Period: 07 October 2023 - 06 October 2030	Verified Carbon Standard	LGAI Technological Center, S.A. (Applus+ Certification)	*Seven years*

1.3 Sectoral Scope and Project Type

Sectoral scope[1]	The project falls under Sectoral Scope 13 - Waste handling and disposal & Sectoral Scope 1 - Energy (renewable/non-renewable)
Project activity type	Type I – Renewable energy projects and Type III – Other Project activity

1.4 Project Eligibility

1.4.1 General eligibility

The   results in GHG emission reductions included in six Kyoto Protocol greenhouse gases, which are listed under the scope of the VCS Program according to VCS Standard version 4.7.[2]. The scope of VCS programme includes

The Seven Kyoto Protocol Greenhouse gases	The project activity is expected to avoid two greenhouse gases[3]: i. Methane (CH_4) emissions from the municipal solid waste (MSW) disposal site in the baseline scenario, which will be avoided in the project scenario.

[1] *Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes:* *https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes*

[2] https://verra.org/wp-content/uploads/2024/04/VCS-Standard-v4.7-FINAL-4.15.24.pdf

[3] https://unfccc.int/sites/default/files/08_unfccc_kp_ref_manual.pdf

	ii. CO_2 emissions from the production of equivalent amount of electricity replaced by the project activity that would otherwise have been produced by conventional fuel-sourced power plants. Thus, the project activity is applicable under the scope of the VCS Program.
Ozone-depleting substances (ODS)	Not applicable, CH_4 and CO_2 are not Ozone-depleting substances which are being emitted from the project activity.
Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	Project activity has been supported by CDM approved methodologies "AMS-III.E. version 17.0 – "Avoidance of methane production from decay of biomass through controlled combustion, gasification or mechanical/thermal treatment"" and AMS-I.D. version 18.0 - "Grid connected renewable electricity generation", and hence, the said condition is not applicable.
Project activities supported by a methodology approved under an approved GHG program, unless explicitly exclude.	The applied methodologies "AMS-III.E. version 17.0 – "Avoidance of methane production from decay of biomass through controlled combustion, gasification or mechanical/thermal treatment"" and AMS-I.D. version 18.0 - "Grid connected renewable electricity generation", is approved under CDM Program, which is a VCS approved GHG program.
Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements.	Not applicable as the project activity is waste to energy plant.

The project activity meets requirements related to the pipeline listing deadline, the opening meeting with the validation/verification body, and the validation deadline as per the table provided below.

As per section 3.8.1 of VCS Standard version 4.7, non-AFOLU projects shall complete validation within two years of the project start date.	The start date of this project activity is 07-October-2023 and the validation is expected to be completed within two years of project start date.

As per section 3.8.7 of VCS Standard version 4.7, a non-AFOLU project applying a standardized method for determining additionality shall initiate the project pipeline listing process within two years of the project start date and may complete validation any time up to concurrent with the first verification	The start date of this project activity is 07-October-2023 on which the plant has commenced to generate electricity through incineration of RDF from the collected waste. The project activity has been submitted for pipeline listing on 21-November-2023 which is within two years of the project start date.
As per section 4.1.5 of VCS Standard version 4.7, The validation/verification body shall ensure that the project is listed on the project pipeline with a status of under validation before the opening meeting with the project proponent, such opening meeting representing the beginning of the validation process. Further, validation shall not begin until the 30-day public comment period has begun, and the validation/verification body shall not complete validation until after the 30-day public comment period has ended.	The project activity was listed on the project pipeline with a status of under validation on 21-November-2023 and the public comment period had ended on 21-December-2023. The opening meeting representing the beginning of the validation process was held on 15-March-2024 which is after the public comment period.

The project uses AMS-III.E.– "Avoidance of methane production from decay of biomass through controlled combustion, gasification or mechanical/thermal treatment" version. 17.0 and AMS-I.D. "Grid connected renewable electricity generation", version. 18.0. The methodology AMS-III.E. applies to project activity involving the prevention of decay of the wastes through controlled combustion; or gasification to produce syngas/producer gas; or mechanical/thermal treatment to produce refuse-derived fuel (RDF) or stabilized biomass (SB) and the methodology AMS-I.D. applies to project activities that comprises renewable energy generation units, such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass, that is supplying electricity to a national or a regional grid.

Since AMS-III.E and AMS-I. D are approved methodology under an approved GHG program, the project is eligible under the scope of the VCS Program according to VCS Standard.

Furthermore, the project activity is not included in any of the excluded project activities, those listed under the Table 1 of VCS Standard ver. 4.7

Hence the project activity is eligible to register under the VCS program standard version 4.7.

1.4.2 AFOLU project eligibility

Not Applicable as the project activity is a non-AFOLU project.

1.4.3 Transfer project eligibility

Not Applicable as the project is not a transfer project.

1.5 Project Design

Indicate if the project has been designed as:

☒ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☐ Grouped project

1.5.1 Grouped project design

Not Applicable, as it is a single location project.

1.6 Project Proponent

Organization name	Antony Lara Renewable Energy Pvt. Ltd
Contact person	Mr.Jose Jacob Kallarakal
Title	Director
Address	Row House No. 6, Kanakia Spaces, Teen Hath Naka, Near TATA Motors Thane(W) 400604
Telephone	0224-2130300
Email	compliance@antonywaste.in

1.7 Other Entities Involved in the Project

Organization name	Kosher Climate India Private Limited
Role in the project	Project representative
Contact person	Mr. Narendra Kumar Ramaraj
Title	Technical Director
Address	11-15/13, Srinivas Nagar Colony, Nagaram, ECIL, Hyderabad, Telangana - 500083, India.

Telephone	+91-9632803444
Email	narendra@kosherclimate.com

1.8 Ownership

Antony Lara Renewable Energy Pvt Ltd is the project proponent of Solid Waste to Energy Plant. They have the legal right to control and operate the project activities.

The Concession agreement, Conversion Charges Agreement, Service & Supply agreement and other documents are evidences for the ownership of the plant, equipment and process that generates GHG emission reductions.

1.9 Project Start Date

Project start date	07-October-2023
Justification	It is the date on which the MSW processing facility started generating electricity through the combustion of RDF processed from the collected MSW and when actual GHG reduction activity started.

1.10 Project Crediting Period

Crediting period	☒ *Seven years, twice renewable* ☐ *Ten years, fixed* ☐ *Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)*
Start and end date of first or fixed crediting period	First Crediting Period (*07 -October-2023 to 06–October-2030*)

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO_2e/year (project)

☐ ≥ 300,000 tCO_2e/year (large project)

Complete the table below for the first (if renewable) or fixed crediting period:

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO_2e)
07 -October-2023 to 31- December-2023	4,374
01 -January-2024 to 31- December-2024	22,685
01 -January-2025 to 31- December-2025	39,681
01 -January-2026 to 31- December-2026	55,788
01 -January-2027 to 31- December-2027	70,992
01 -January-2028 to 31- December-2028	85,550
01 -January-2029 to 31- December-2029	99,191
01 -January-2030 to 06- October-2030	83,495
Total estimated ERRs during the first or fixed crediting period	461,756
Total number of years	07
Average annual ERRs	57,719

1.12 Description of the Project Activity

Prior to the implementation of the project activity, MSW in Pimpri-Chinchwad was mainly subjected to landfill in anaerobic condition without any methane recovery system for flaring or energy recovery purpose, therefore in the absence of the project activity, methane emissions from anaerobic decay of waste would have been released into the atmosphere.

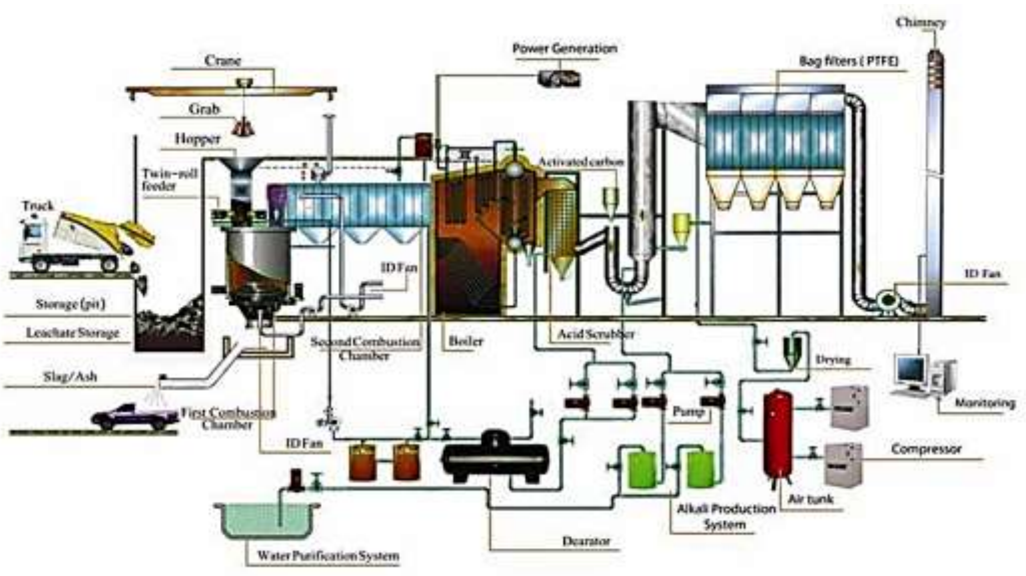
As per para. 19 of AMS-I. D, “Grid connected renewable electricity generation the baseline scenario is that the electricity”, Version 18 delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

The project activity involves R grate incineration technology to incinerate RDF prepared from MSW sourced from Pimpri-Chinchwad city that would otherwise be disposed of at a local landfill site. The main equipment in the project activity includes 1 incinerator (700 ton of RDF per day), 1 boiler (70 tons per hour), 1 steam turbine and 1 generator (14 MW), the transformer station and electricity transmission line, exhaust gas cleaning systems and ash removal systems.

The steam generating system of the Project activity consist of a Municipal Solid Waste (MSW) fired incinerator boiler with a Maximum Continuous Rating (MCR) of 1 x 70 TPH, with the outlet steam parameters at 45 kg/cm2g and 410°C. The control range on the super-heater outlet temperature shall be ±5ºC. The combustion system of the boiler is stoker type. The boiler efficiency when firing 100% MSW, will be a minimum of 68 % on the Gross Calorific Value (GCV) basis. The dust concentration in the flue gases leaving the boiler will be a maximum of 50 mg/Nm³. The grate acts as a continuous ash discharge and an overfeed stoker, with hydraulic drive. Fuel is burnt, suspended in air above the grate and on the traveling grate. Ash generated in the boiler is continuously discharged. Air seals are provided in order to control the excess air entry in to the furnace.

The lifetime of the project activity is 21 years based on the concession period and the technical specifications of major equipment are provided in table below.

The layout of the plant is shown in below figure:

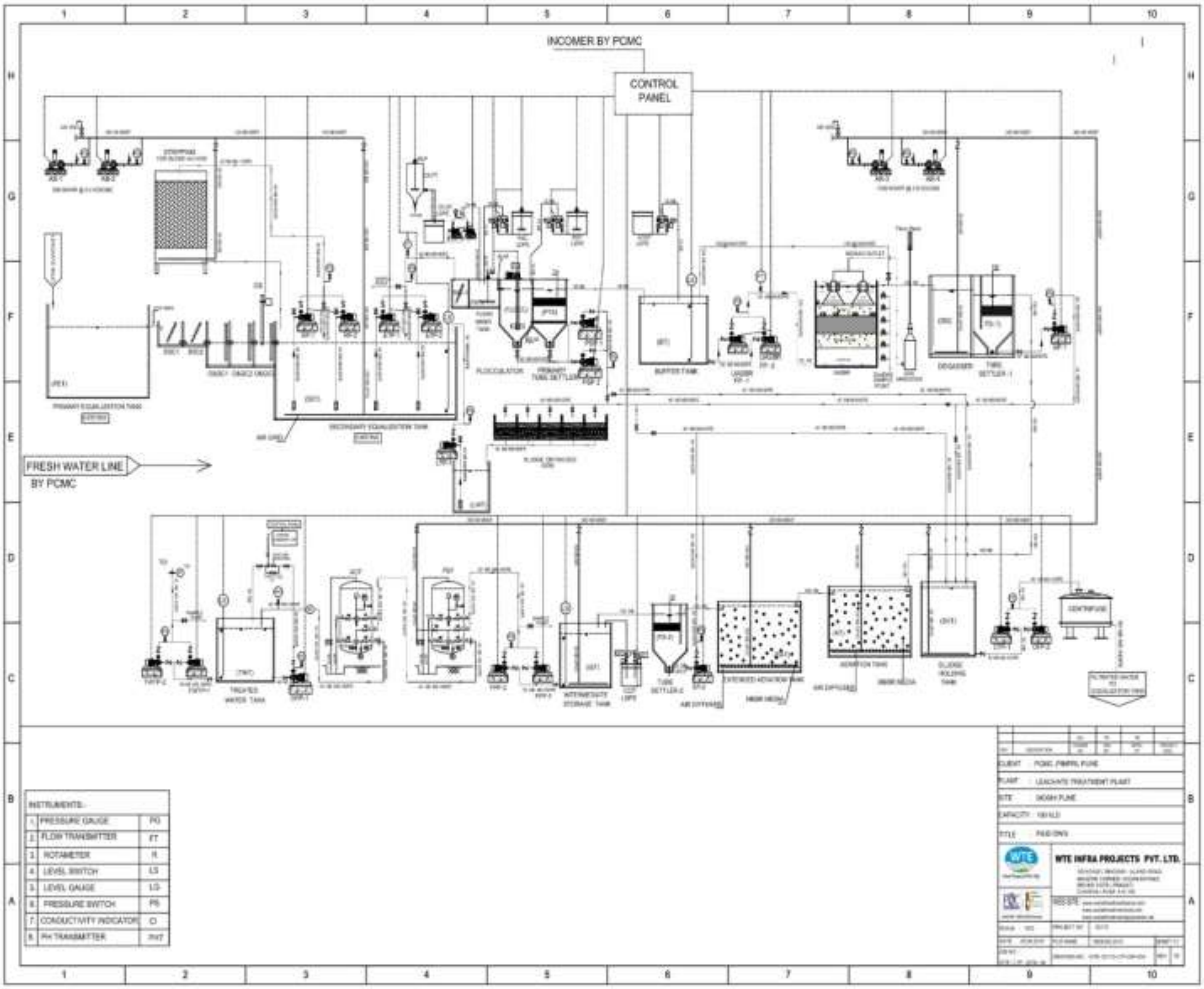


The main technical details of project is shown in table below:

Main Component	Indicator	Values
Incinerator/ Furnace	Type	R Grate Type
	Manufacturer	ISGEC Heavy Engineering Ltd
	Number	1
	Unit capacity	700 ton/day
	Lifetime	25 years
Boiler	Number	1
	Manufacturer	Hitachi Zosen India Pvt. Ltd. (HZIPL)
	Unit steam output	66.5 ton/h
	Rated Outlet Steam Pressure	46 Bar

	Rated Steam Temperature	410 ^{o}C
	Max allowed Steam temperature	415 ^{o}C
	Lifetime	25 years
Steam turbine	Type	Condensing turbine (Bleed Cum)
	Number	1
	Rated capacity	14 MW
	Speed	153 rpm
	Input steam	400^{o}C
	Steam Pressure	44.5 BAR (A)
	Lifetime	25 years
Generator	Number	1
	Capacity	14 MW
	Frequency	50 Hz +5%
	Rated voltage	11 kV
	Rated speed	1500 rpm
	Lifetime	25 years

The leachate discharged from the bunker and the landfill site have been collected through drains and sent to leachate treatment plant of PCMC and treated through several steps as per the flow diagram given below to meet the environmental quality standard.



1.13 Project Location

The Project is located in PCMC Garbage depot, Tapkir Nagar, Moshi Pimpri Chinchwad-412105, Pune district, Maharashtra, India. The geo-coordinate of the project is 18°39'29.52"N, 73°51'26.51"E. The location map is given below.

1.14 Conditions Prior to Project Initiation

The scenario existing prior to the start of the implementation of the project activity is:

Waste Treatment:

In the absence of the project activity, the municipal solid waste generated in Pimpri-Chinchwad city were subjected to anaerobically managed landfill, where the waste was left to decay in anaerobic conditions. The greenhouse gases (GHG) generated from the solid waste were released directly into atmosphere.

Electricity Generation:

In the absence of the project activity, the electricity supply for Pimpri-Chinchwad city was generated mainly from grid connected power plants, where the use of fossil fuel would also lead to GHG emissions.

The baseline scenario is the same as the scenario existing prior to the start of the implementation of the project activity. Please refer to Section 3.4 (Baseline Scenario) for details.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project activity does not fall under the purview of the Environmental Impact Assessment (EIA) notification of the Ministry of Environment, Forest & Climate Change (MOEF&CC), Government of India (As per MOEF&CC Notification No. - S.O. 1533, dated 14th September 2006[4]). In addition to that, all applicable laws and regulations in India were complied with.

The project activity is implemented in compliance to The Factories Act, 1948[5] & The Indian Boilers Act-1923[6].

The project proponent has obtained approval from Maharashtra Pollution Control Board (MPCB) to set up the Municipal Solid Waste based Power Project.

The project activity is implemented in compliance to The Factories Act, 1948[7] & The Indian Boilers Act-1923[8].

The project activity is in compliance with the Municipal Solid Waste (Management and Handling) Rules 2000[9], where all the municipal authorities in country are directed to manage solid waste in their respective jurisdiction. The central government has revised and replaced the MSW Rules 2000 with Solid Waste Management (SWM) Rules 2016[10] & Solid Waste Management (SWM) Rules 2020[11] (amended). The new rules are applicable beyond municipal areas and include urban agglomerations, census towns, notified industrial townships, areas under the control of Indian Railways, airports, special economic zones, places of pilgrimage, religious and historical importance, and State and Central Government organisations in their ambit. The key areas covered under these rules are segregation at source, collection & Disposal of sanitary waste, collect back mechanism for packaging waste, user fee for collection, waste processing and treatment, promotion of compost, promotion of waste to energy, revision of parameters and standards etc. The project activity is in compliance with the National Urban Sanitation Policy,

[4] https://environmentclearance.nic.in/writereaddata/EIA_Notifications/1_SO1533E_14092006.pdf

[5] https://environmentclearance.nic.in/report/EIA_Notifications.aspx

[6] https://www.ilo.org/dyn/natlex/docs/ELECTRONIC/48110/114241/F-1729379182/IND48110.pdf

[7] https://environmentclearance.nic.in/report/EIA_Notifications.aspx

[8] https://www.ilo.org/dyn/natlex/docs/ELECTRONIC/48110/114241/F-1729379182/IND48110.pdf

[9] https://cpcb.nic.in/uploads/MSW/MSW_AnnualReport_2001-02.pdf

[10] https://cpcb.nic.in/uploads/MSW/SWM_2016.pdf

[11] https://qps.nhsrcindia.org/sites/default/files/2022-01/Solid%20Waste%20Management%20%28Amendment%29%20Rule%202020.pdf

2008[12], that has the goal to transform Urban India into community-driven, totally sanitized, healthy and liveable cities and towns. The project activity is in compliance with the plastic waste management rules, 2016[13].

The Swachh Bharat Mission or Clean India Mission, 2014 was launched to eliminate the open defecation, eradication of manual scavenging, modern and scientific municipal solid waste management. The project activity is in compliance with the Swachh Bharat Mission and AMRUT.

The project activity is in compliance with all relevant statutory and regulatory laws applicable in the host country.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

[12] https://www.mohua.gov.in/upload/uploadfiles/files/NUSP_0.pdf

[13] https://thc.nic.in/Central%20Governmental%20Rules/Plastic%20Waste%20Management%20Rules,%202016.pdf

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ *No*

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☒ No

1.18 Sustainable Development Contributions

This plant operation directly covers key areas of sustainable development including renewable power generation, solid waste management, wastewater management and efficient usage, environment protection, innovative technological integrations, and partnership for sustained growth. The MSW based waste to energy plant has incorporated many of the Sustainable Development Goals (SDGs) of United Nations.

Among 17 SDGs, 7 SDGs are directly linked with the design of the campus and process innovation of this MSW to renewable energy production facility:

1. Clean water and sanitation (SDG-6)

Goal: Ensure availability and sustainable management of water and sanitation for all

Target: 6.3 By 2030, improve water quality by reducing pollution, eliminating dumping and minimizing release of hazardous chemicals and materials, halving the proportion of untreated wastewater and substantially increasing recycling and safe reuse globally.

Indicator: 6.3.1 Proportion of domestic and industrial wastewater flows safely treated

ALREPL shall employ all water conservation methods to ensure that the plant does not consume any fresh water (Zero Fresh Water Intake plant).

Monitoring Parameter: Net Inlet Quantity of sewage treated water per day (100m^3/day)

2. Affordable and clean energy (SDG-7)

Goal: Ensure access to affordable, reliable, sustainable and modern energy for all

Target: 7.2: By 2030 increase substantially the share of renewable energy in the global energy mix

Indicator: 7.2.1: Renewable energy share in the total final energy consumption

The plant capacity of 14MW generates renewable power. To calculate the carbon emission that would have otherwise generated from fossil fuel energy generation and also to substantiate SDG 7.

Net quantity of renewable energy generated from the power plant, which otherwise would have been generated from the grid connected power plants.

Monitoring Parameter: Net Electricity Generation (GWh/year)

3. Decent work & economic growth (SDG-8)

Goal: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

Target: 8.5 By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value

Indicator: 8.5.1 Average hourly earnings of employees, by sex, age, occupation and persons with disabilities

The project activity provides employment opportunities and the average salaries has been paid to employees irrespective of sex, age group, etc which leads to increased productivity, driving economic development. By additional revenue generation from municipal waste, post combustion ash and recovering valuable metals and non- metals from the waste.

Monitoring Parameter: No of employment (with bifurcation on number by sex, age group and where applicable, persons with disabilities), Average earnings, Policy for Non-discrimination and equal pay for the work of equal value.

4. **Sustainable cities and communities (SDG-11)**

Goal: Make cities and human settlements inclusive, safe, resilient and sustainable

Target: 11.6 By 2030, reduce the adverse per capita environmental impact of cities, including by paying special attention to air quality and municipal and other waste management

Indicator: 11.6.1 Proportion of municipal solid waste collected and managed in controlled facilities out of total municipal waste generated, by cities

Monitoring Parameter: Net Inlet Quantity of Municipal Solid Waste (Tonnes per day)

5. **Responsible consumption and production (SDG-12)**

Goal: Ensure sustainable consumption and production patterns

Target: 12.5 By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse

By adopting material conservation and waste recycling approaches in plant design and operations. Recovery of metal and non-metal waste from MSW for re-using to boost circular economy.

Monitoring Parameter: Net Inlet Quantity of Municipal Solid Waste (Tonnes per day), Quantity of scrap metal recovered (Tonnes per day)

6. **Climate action (SDG-13)**

Goal: Take urgent action to combat climate change and its impacts

Target: 13.2 Integrate climate change measures into national policies, strategies and planning

Indicator: 13.2.2 Total greenhouse gas emissions per year

Reduction of annual average of 57,719 tons of CO_2 eq. greenhouse gas (GHG) emissions per year by coal replacement and avoiding open dumping of waste.

Project activity generates renewable energy-based electricity and mitigates the CO_2 emissions which would have been generated from the grid-connected power plants. Project has already commissioned and started reducing the emissions. Hence, complied to the SDG 13

Monitoring Parameter: GHG emission reductions (tCO_2/year)

7. **Partnership for the goals (SDG-17)**

Goal: Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development

Target: 17.17 Encourage and promote effective public, public-private and civil society partnerships, building on the experience and resourcing strategies of partnerships

By setting up of WTE plant in PPP (Public-Private Partnership) between M/s Antony Lara Renewable Energy Pvt. Ltd (ALREPL) and PCMC (Pimpri Chinchwad Municipal Corporation).

1.19 Additional Information Relevant to the Project

Leakage Management

As per methodology AMS III E, version 17.0 para 37 under leakage emissions, “In case of RDF/SB production, project proponents shall demonstrate that the produced RDF/SB is not subject to anaerobic conditions before its combustion end-use resulting in methane emissions”. As per real-scenario, the entire waste received in a day will be processed and treated on the same day. This ensures that the waste received in the plant is treated on a FIFO basis. Handling and storage of the collected RDF in the bunker does not undergo anaerobic condition as the continuous mixing of waste takes place to avoid moisture absorption and maintain homogeneity. In case of plant breakdown, the accumulated waste will be stored in enclosed bunker provided with adequate ventilation to promote aerobic conditions and the accumulated waste prior to current operation day will be scheduled for incineration to further avoid any anaerobic degradation. This ensures produced RDF/SB does not result in anaerobic conditions and do not lead to further absorption of moisture. Hence, no leakage emission is considered in the project activity.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not Applicable

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The stakeholders for the project have been prioritized by identifying direct (those who have a direct interest or influence on the project) and indirect stakeholders whose interest is indirect. Stakeholders has been identified based on potential impact from all related facilities associated with the project including associated facilities, transport routes etc. Hence, the stakeholders identified are Local people, employees, Local authorities, NGOs, drivers, etc.
Legal or customary tenure/access rights	The legal and customary rights of the land during and post implementation of the project lies with Pimpri Chinchwad Municipal Corporation. In line with the concession agreement, the license fee for the land utilized for the project activity will be paid to Pimpri Chinchwad Municipal Corporation throughout the concession period of 21 years. Project proponent have no rights on the legal ownership of the land. There are no identified disputes or conflicts over rights of the land and its resources. No concerns were received against the implementation of the project activity and confirms the consent and full support for the project implementation.
Stakeholder diversity and changes over time	Each community has its own cultural practices, customs, and beliefs, contributing to the overall cultural heritage and integrity of the region. The project anticipates no significant changes to the group's cultural integrity in the project region. Additionally, project creates additional livelihood promotion among the local communities helps to improve the sanitary conditions paving way for well-being in the project region. Further, the project activity by contributing to the

	environmental, economic, and social benefits, play a crucial role in enabling local communities to remain located in the project area.
Expected changes in well-being	As the project leads to avoidance of methane emissions through proper disposal of Municipal solid waste and generation of electricity, the baseline would have involved emission of methane from the landfill site and generation electricity from the grid connected plants, the project activity enhances the well-being of the local communities in the aspects such as sanitation, socio-economic performance, etc
Location of stakeholders	All community members, those who are directly or indirectly impacted by the project are mostly local people who reside near the project areas.
Location of resources	The land on which the project is implemented, and its resources are owned and managed by the Pimpri Chinchwad Municipal Corporation. All land and resource rights including the access and customary rights vests with the Pimpri Chinchwad Municipal Corporation who reside in the project study area.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	Stakeholder Consultation 1 – 09-August-2021 Stakeholder Consultation 2 - 03-November -2023
Stakeholder engagement process	**Stakeholder Consultation 1** As a part of Environmental and Social Impact Assessment, a local stakeholder engagement was held on 09-August-2021 for which the PCMC authority and the local people were participated. **Stakeholder Consultation 2** The LSC invitations were sent to Commissioner office on 25-October-2023 and Maharashtra Pollution Control Board on 31- October-2023 and the local people were also invited. The consultation attendees were given sufficient time and scope to discuss the project and clarify their doubts related to the project and provide their feedbacks on the project.

Consultation outcome	The PP has conducted two discussion meetings on the project particulars with stakeholders. These sessions ensured unbiased dissemination of all project-related information to stakeholders, irrespective of caste, religion, gender, race, etc., This approach aimed to eliminate any potential barriers hindering the participation of diverse stakeholder groups. The first LSC discussion covered topics, including project process, objectives, project benefits, risks along with consideration to register the project under a suitable carbon mechanism for availing the carbon credits and its financial benefits. As the first LSC meeting was held with limited scope on the carbon project cycle including its processes, a second LSC discussion was held covering a wide range of topics, including project design, location, objectives, costs, benefits, risks, duration, scope, size, impacts, and the carbon project cycle, encompassing validation and verification processes. Based on this information, interested stakeholders provided their inputs through open discussions during the meetings. No negative comments have been received on project activity from any of the local stakeholders consulted. As all comments were very positive about the project, no further action is required. There were no further comments raised by the stakeholders and they were totally in support for setting up of these kinds of projects in the region.
Ongoing communication	A dedicated email address as well as the phone number and the address of Antony Lara Renewable Energy Pvt. Ltd were provided to all stakeholders to foster open and accessible communication. There will be an active presence of the Antony Lara Renewable Energy Pvt. Ltd team in the project area. They will work closely with the community and other stakeholders to address any queries or grievances in relation to the project. All communication and consultation shall be performed in a culturally appropriate manner, including language and gender sensitivity.
Stakeholder input	No negative comments have been received on project activity from any of the local stakeholders consulted. As all comments were very positive about the project, no further action is required.

2.1.3 Free Prior and Informed Consent

Obtaining consent	The legal and customary rights of the land during and post implementation of the project activity lies with Pimpri Chinchwad Municipal Corporation. In line with the concession agreement, the license fee for the land utilized for the project activity will be paid to Pimpri Chinchwad Municipal Corporation throughout the concession period of 21 years. Project proponent have no rights on the legal ownership of the land. There are no identified disputes or conflicts over rights of the land and its resources. No concerns were received against the implementation of the project activity and confirms the consent and full support for the project implementation.
Outcome of FPIC	As per the concession agreement with the Pimpri Chinchwad Municipal Corporation, the project proponent has the legal right to implement the project activity as well as it has the legal rights on the GHG emission reductions generated (carbon credits) from the project activity. Hence the PP has not encroached on land or relocated people.

2.1.4 Grievance Redress Procedure

Development process	The project team devised a structured process in a culturally appropriate manner for receiving input and addressing grievances, through initial consultations with stakeholders. After being accepted by the majority without further input, the procedure was finalized.
Grievance redress procedure	There is a mechanism in place to address the grievances and resolve any potential conflict between the project proponent and local stakeholders. The project proponent contact information like e-mail and phone numbers are provided to the stakeholders in case they want to contact and share their ideas and convey their grievances.

2.1.5 Public Comments

This project was open for public comment from 21-November-2023 to 21-December-2023. No comments were received during and subsequent to the public comment period.

Comments received	Actions taken
Not applicable as No comments were received during and subsequent to the public comment period.	Not applicable as no comments were received during and subsequent to the public comment period.

2.2 Risks to Stakeholders and the Environment

	Risks identified	Mitigation or preventative measure taken
Risks to stakeholder participation	No risks identified	The project team has explained the project and project activities to the stakeholders including project benefits, costs, risks, rights, laws and opportunities on employment generation. The stakeholders are voluntarily willing to participate in the project because of benefits like safeguarding the communities from environmental challenges and various other SDG benefits of the project.
Working conditions	No risks identified	The project has taken comprehensive measures to ensure providing safe and healthy working conditions for its employees by conducting safety trainings twice in a month and measures to maintain hygiene and odour control has been carried out at the project site.
Safety of women and girls	No risks identified	The women are provided employment opportunities in the project activity with attention given to addressing gender-specific safety concerns. The organization does not tolerate harassment for any women employees which includes sexual, psychological or any other form of

		harassment that can be a form of discrimination.
Safety of minority and marginalized groups, including children	No risks identified	The project proponent has protocols aimed at safeguarding women, children and other marginal or vulnerable groups.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	No risks identified	Although The project activity does not involve any significant generation of pollutant, but in case of natural calamities or fire accidents, the project activity shall be subjected to generation of pollutants.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Discrimination and sexual harassment	The organization has policies for Discrimination and sexual harassment and would take appropriate steps for the prevention of such activities at work workplace and put in place an appropriate mechanism of complaint/grievance redressal.
Management experience	The Project proponent has been developing and operating environmentally safe waste disposal facilities, and has continually provided the most up-to-date environmental solutions for non-hazardous waste reduction and disposal. And has the operational expertise, management strengths, financial capabilities and commitment to quality and the environments.
Gender equity in labor and work	The women have been given job opportunities in the project site and the HR policy extends to all the employees of the company irrespective of gender.
Human trafficking, forced labor, and child labor	The HR policy ensures the prevention of instances of Harassment in the workplace, human trafficking, forced labor, and child labor.

2.3.2 Human Rights

The PP emphasize on aligning the organizational practices with the human rights outlined in national and regional regulations. All customary rights to the land and its resources are held by the respective Municipal Corporation. The project respects protection of the rights of local communities and customary rights holders, aligning with applicable international human rights law, the United Nations Declaration.

2.3.3 Indigenous Peoples and Cultural Heritage

The project does not impact cultural heritage as part of project activities.

2.3.4 Property Rights

Rights to territories and resources	**The legal and customary rights of the land during and post implementation of the project activity lies with Pimpri Chinchwad Municipal Corporation. In line with the concession agreement, the license fee for the land utilized for the project activity will be paid to Pimpri Chinchwad Municipal Corporation throughout the concession period of 21 years.** **Project proponent have no rights on the legal ownership of the land.** **There are no identified disputes or conflicts over rights of the land and its resources.**
Respect for property rights	The project respects all stakeholder property rights. All rights except right to carbon generated from project activity, remain with the respective Municipal Corporation. The Project Proponent will not alter or impact any of their rights, thus preserving and protecting them through the concession agreements.

2.3.5 Benefit Sharing

Not Applicable as explained in the table given below:

Process used to design the benefit sharing plan	Not Applicable as the project proponent holds ownership of entire VCUs and there is no benefit sharing involved.
Summary of the benefit sharing plan	Not Applicable as the project proponent holds ownership of entire VCUs and there is no benefit sharing involved.
Approval and dissemination of benefit sharing plan	Not Applicable as the project proponent holds ownership of entire VCUs and there is no benefit sharing involved.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure taken
Impacts on biodiversity and ecosystems	No risk identified	The project does not have any impact on biodiversity and ecosystems as the project land is barren.
Soil degradation and soil erosion	No risk identified	The land use of project is barren which is not suitable for agriculture.
Water consumption and stress	No risk identified	The project involves water conservation measures to ensure the plant does not consume any fresh water.
Usage of fertilizers	No risk identified	There is no usage of fertilizers involved.

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

2.4.2 Introduction of species

As the project is not a ARR, ALM, WRC or ACoGS project, this section is not applicable.

2.4.3 Ecosystem conversion

As the project is not a ARR, ALM, WRC or ACoGS project, this section is not applicable.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The project uses two approved CDM small-scale methodologies

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS III.E	"Avoidance of methane production from decay of biomass through controlled combustion, gasification or mechanical/thermal treatment"[14]	17.0
Methodology	AMS I.D	Grid connected renewable electricity generation[15]	18.0
Tool	Tool 03	Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion[16]	03.0
Tool	Tool 04	Tool to calculate emissions from solid waste disposal sites[17]	08.1
Tool	Tool 07	Tool to calculate the emission factor for an electricity system[18]	07.0
Tool	Tool 12	Project and leakage emissions from transportation of freight[19]	01.1.0

[14] https://cdm.unfccc.int/UserManagement/FileStorage/YUCF7O03GWKZEQBRXVTA5H4DJLMI9S

[15] https://cdm.unfccc.int/UserManagement/FileStorage/2P7FS6ZQAR84LG3NMKYUH50WI9ODBC

[16] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-03-v3.pdf

[17] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-04-v8.1.pdf

[18] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf

[19] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-12-v1.1.0.pdf

Tool	Tool 21	Demonstration of additionality of small-scale project activities[20]	13.1
Tool	Tool 27	Investment analysis[21]	14.0

3.2 Applicability of Methodology

The applicability criteria of the applied methodology are demonstrated below.

Methodology ID	Applicability condition	Justification of compliance
Reference Methodology applied: AMS III.E, ver. 17.0		
1	The project activity does not recover or combust methane unlike AMS-III.G. Nevertheless, the location and characteristics of the disposal site in the baseline condition shall be known, in such a way as to allow the estimation of its methane emissions.	Before the implementation of the project activity, the municipal solid waste was disposed at the PCMC Landfill site involving controlled placement of waste which was directed to specific deposition areas. Subsequent to this process, levelling of the waste was performed. Therefore, it is considered as anaerobically managed solid waste disposal site, as the characteristics is consistent with the definition provided in tool 04, version 08.1. The project activity involves methane avoidance through controlled combustion, gasification or mechanical/thermal treatment of municipal solid waste. There is no methane recovery or combustion process involved. Hence, the project activity meets the applicable condition.

[20] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-21-v13.1.pdf

[21] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v14.0.pdf

2	If the project involves combustion, gasification or mechanical/thermal treatment of partially decayed waste mined (i.e., removed) from a solid waste disposal site in addition to freshly generated waste the project participants shall demonstrate that there is adequate capacity of the combustion, gasification or mechanical/thermal treatment facility to treat the newly generated wastes in addition to the partially decayed wastes removed from the disposal site. Alternately justifications for combusting, gasifying or mechanically/thermally treating the partially decayed wastes instead of the newly generated wastes shall be provided.	The project activity uses fresh waste sourced from Pimpri-Chinchwad Municipal Corporation and there is sufficient capacity available for the utilization of waste to produce electricity. Hence the project meets the applicability condition.
3	If the combustion facility, the produced syngas, producer gas or RDF/SB is used for heat and electricity generation within the project boundary, that component of the project activity may use a corresponding methodology under Type I project activities.	The project involves utilisation of 700 TPD RDF (pre-processed/segregated waste) and generation of electricity with the installed capacity of 14 MW. The electricity generated is supplied to the Indian grid. The relevant methodology AMS- I.D is applied to the project activity. Hence the condition is applicable to the project activity.
4	In case of RDF/SB production, project proponents shall provide evidence that no GHG emissions occur, other than biogenic CO_2, due to chemical reactions during the thermal treatment process for example limiting the temperature of thermal treatment to prevent the occurrence of pyrolysis and/or the stack gas analysis.	Incineration of municipal solid waste (MSW) generally results in negligible emissions of greenhouse gases like nitrous oxide and Carbon monoxide which has been evidenced from the Continuous Emission Monitoring System (CEMS) records due to the presence of nitrogen and carbon compounds in the waste. It is also to be noted from the CEMS records that the flue gas emissions are found to be

		within the CPCB flue gas emission standards[22] However, the project activity involves processing of RDF pre-dominantly with organic content and higher combustible properties which has been supported by physical waste characterization report. Hence, the project activity majorly involves generation of biogenic CO_2. Additionally, as per the methodology AMS. III. E, version 17.0, the emissions from fossil carbon content have been accounted in the project emissions from combustion of RDF. Hence, the project meets the applicability condition.
5	In case of gasification, the process shall ensure that all the syngas produced, which may contain non-CO_2 GHG, will be combusted and not released unburned to the atmosphere. Measures to avoid physical leakage of the syngas between the gasification and combustion sites shall also be adopted.	The project activity does not involve gasification process. Hence this condition is not applicable.
6	In case of RDF/SB processing, the produced RDF/SB should not be stored in such a manner as resulting in high moisture and low aeration favouring anaerobic decay. Project participants shall provide documentation showing that further handling and storage of the produced RDF/SB does not result in anaerobic conditions and do not lead to further absorption of moisture.	The entire waste received in a day will be processed and treated on the same day. This ensures that the waste received in the plant is treated on a FIFO basis. Handling and storage of the collected RDF in the bunker does not undergo anaerobic condition as the continuous mixing of waste takes place to avoid moisture absorption and maintain homogeneity. In case of plant breakdown, the accumulated waste will be stored in enclosed bunker provided with adequate ventilation to promote aerobic

[22] https://www.cpcb.nic.in/common-hw-incinerators-annexure/

		conditions and the accumulated waste prior to current operation day will be scheduled for incineration to further avoid any anaerobic degradation. This ensures produced RDF/SB does not result in anaerobic conditions and do not lead to further absorption of moisture.
7	In case of RDF/SB processing, local regulations do not constrain the establishment of RDF/SB production plants/thermal treatment plants nor the use of RDF/SB as fuel or raw material.	There are no local regulations that constrain the production/ utilisation of RDF. However, the project proponent has obtained necessary consent from Maharashtra Pollution Control Board (MPCB) for the establishment of waste to energy plant. Hence this criterion is not applicable.
8	During the mechanical/thermal treatment to produce RDF/SB no chemical or other additives shall be used.	The RDF is produced from the municipal solid waste sourced from Pimpri-Chinchwad city. The waste is pre-processed to remove the unwanted/inert materials. During the mechanical/thermal treatment, no chemical or additives is used to produce RDF. Hence this criterion is not applicable.
9	In case residual waste from controlled combustion, gasification or mechanical/thermal is stored under anaerobic conditions and/or delivered to a landfill, emission from the residual waste shall to be taken into account using the first order decay model (FOD) described in AMS-III E	The residual waste from the controlled combustion process includes bottom ash and fly ash. The fly ash and bottom ash will be stored and sent to Construction & Demolition waste processing plant for further production of construction materials. Therefore, it is not subjected to anaerobic storage. Hence this condition is not applicable.
Title: Grid connected renewable electricity generation		
Reference of methodology applied: AMS I.D, ver. 18.0		
1	This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s);	The project activity is a greenfield, Indian grid-connected renewable waste to energy plant. Therefore, it confirms to the said criteria.

	(c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or Involve a replacement of (an) existing plant(s).	
2	Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m^2; The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m^2.	The project activity is installation of new grid-connected waste to energy plant. Therefore, the said criterion is not applicable.
3	If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	The project activity is the installation of a new grid-connected waste to energy plant. Hence this criterion is not applicable
4	Combined heat and power (co-generation) systems are not eligible under this category.	The project activity is the installation of a new grid connected waste to energy plant which does not involve co-generation. Hence this criterion is not applicable

5	In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	The project activity is the installation of a new grid connected waste to energy plant. Hence this criterion is not applicable
6	In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW.	The project activity is the installation of a new grid connected waste to energy plant. Hence this criterion is not applicable.
7	In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid, then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as "AMS-I.C.: Thermal energy production with or without electricity" shall be explored.	The project activity involves treatment of municipal waste generated from Pimpri-Chinchwad area to produce RDF (Refused Derived Fuel). The relevant methodology AMS-III. E. is used for the methane avoidance component through controlled combustion, gasification or mechanical/thermal treatment. There is no recovery of methane from waste. Hence this criterion is not applicable.
8	In case biomass is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass" shall apply.	The project activity involves treatment of municipal waste generated from Pimpri-Chinchwad area to produce RDF (Refused Derived Fuel). Hence this criterion is not applicable.
Tool 03 Title: Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion, Version 03.0		
1.	This tool provides procedures to calculate project and/or leakage CO_2	The project involves consumption of diesel for the operation of waste handling vehicle

	emissions from the combustion of fossil fuels. It can be used in cases where CO_2 emissions from fossil fuel combustion are calculated based on the quantity of fuel combusted and its properties. Methodologies using this tool should specify to which combustion process j this tool is being applied.	and loaders, and for the DG sets. Hence the project activity meets the applicable condition.
Tool 04, Title: Tool to calculate emissions from solid waste disposal sites, Version 08.1		
1	The tool can be used to determine emissions for the following types of applications Application A: The CDM project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by capturing and flaring or combusting the methane (e.g., "ACM0001: Flaring or use of landfill gas"). The methane is generated from waste disposed in the past, including prior to the start of the CDM project activity. In these cases, the tool is only applied for an ex-ante estimation of emissions in the project design document (CDM-PDD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e.g., measuring the amount of methane captured from the SWDS); Application B: The CDM project activity avoids or involves the disposal of waste at a SWDS. An example of this application of the tool is ACM0022, in which municipal solid waste (MSW) is treated with an alternative option, such as composting or anaerobic digestion,	The project activity involves treatment of municipal waste generated from Pimpri-Chinchwad area to produce RDF (Refused Derived Fuel). The methane emissions from waste disposed is avoided during the crediting period through controlled combustion of RDF. Hence the condition B is applicable to the project activity.

	and is then prevented from being disposed of in a SWDS. The methane is generated from waste disposed or avoided from disposal during the crediting period. In these cases, the tool can be applied for both ex-ante and ex post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions.	
Tool 07 Title: Tool to calculate the emission factor for an electricity system, Version 07.0		
1	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects).	The project activity is a greenfield waste to energy generation plant and hence, according to the applied methodology, the baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in "TOOL07: Tool to calculate the emission factor for an electricity system", version 07.0.0.
2	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e., option II a and option IIb. If option II a is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be	Since the project activity is grid connected waste to energy project this condition is applicable. Combined margin grid emission factor has been calculated as per the CO_2 emission factor data base, version 19.0 published by the CEA[23] in which for the calculation of emission factor CEA have only considered grid connected plants.

[23] https://cea.nic.in/cdm-co2-baseline-database/?lang=en

	at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
Tool 12 Title: Project and leakage emissions from transportation of freight, Version 01.1.0		
1.	This tool is applicable to project activities which involve freight transportation by road and where transportation is not the main project activity. This tool is not applicable to project activities where transportation is the main source of greenhouse gases emissions. This tool does not provide procedures to estimate baseline emissions from road transportation of freight. The tool only provides to determine CO_2 emissions. CH_4 and N_2O emissions are excluded for simplification as they are small compared to CO_2 emissions.	The project activity is a greenfield waste to energy generation plant. The project activity involves transportation of Municipal Solid Waste (MSW) from the waste collection point to the project site. The transportation of MSW is not the main project activity. Hence, this tool is applicable.
2.	In addition, the tool is applicable for the determination of project or leakage emissions from freight transportation by rail in project activities where transportation is not the main project activity.	The project activity involves estimation of project emissions from transportation of freight by road. Hence this condition is not applicable.
Tool 21 Title: Demonstration of additionality of small-scale project activities, Version 13.1		
1	The use of the methodological tool “Demonstration of additionality of small-scale project activities” is not mandatory for project participants	The project proponent is not proposing any new methodology and applied this tool for demonstration of additionality with reference to the applied methodology AMS-

	when proposing new methodologies. Project participants and coordinating/managing entities may propose alternative methods to demonstrate additionality for consideration by the Executive Board.	I. D, Grid connected renewable electricity generation" version 18.0. Refer to section 3.5 of the PD for the detailed applicability of this tool and additionality assessment.
2	Project participants and coordinating/managing entities may also apply "TOOL19: Demonstration of additionality of microscale project activities" as applicable.	This is a small-scale project activity hence TOOL19: "Demonstration of additionality of microscale project activities" is not applicable.
Tool 27 Title: Investment analysis, Version 14.0		
1	This methodological tool is applicable to project activities that apply the methodological tool "Tool for the demonstration and assessment of additionality", the methodological tool "Combined tool to identify the baseline scenario and demonstrate additionality", the guidelines "Non-binding best practice examples to demonstrate additionality for SSC project activities", or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	The project activity applies "non-binding best practice examples to demonstrate additionality for SSC project activities", Hence this tool is applicable.
2	In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The project activity is a small-scale grid connected Waste to Energy power project. It refers to the "non-binding best practice examples to demonstrate additionality for SSC project activities", Hence this tool is applicable.

3.3 Project Boundary

The project boundary defined as per the applied methodologies are:

(i) According to paragraph 21 of AMS-III.E, version 17.0

The project boundary are the physical, geographical sites:

(a) Where the solid waste would have been disposed or is already deposited and the avoided methane emission occurs in absence of the project activity;

(b) Where the treatment of biomass through controlled combustion, gasification or mechanical/thermal treatment takes place;

(c) Where the final residues of the combustion process will be deposited (this parcel is only relevant to controlled combustion activities);

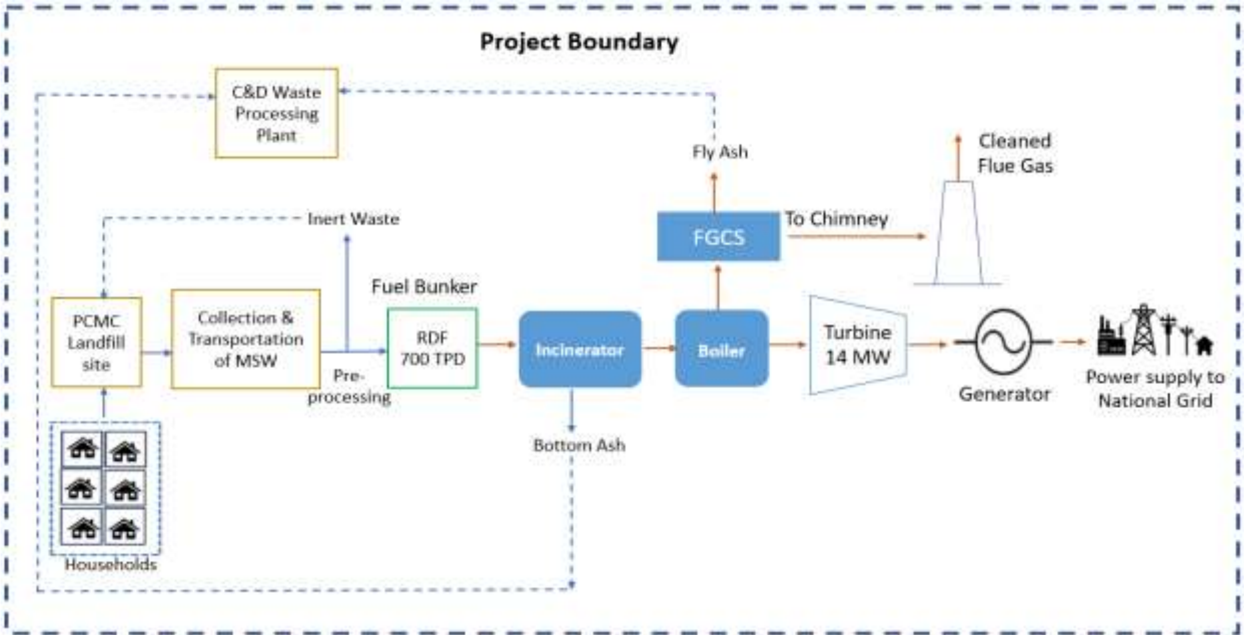
(d) And in the itineraries between them, where the transportation of wastes and combustion residues and/or residues of gasification and mechanical/thermal treatment process occurs.

As per the above-mentioned criteria, the project boundary includes the sites of waste collection, waste transportation from Pimpri-Chinchwad city to plant, the project site, transportation of inert waste to PCMC landfill site and bottom and fly ash which is sent to Construction & Demolition waste processing plant.

According to paragraph 18 of AMS-I. D, version 18.0

The spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the CDM project power plant is connected to.

The project activity involves the incineration of RDF prepared from MSW and exports electricity to the Indian grid. The project boundary includes the site of treatment process, on-site electricity and/or heat generation out of which electricity is supplied to grid and all the other plants connected to the grid. The project boundary is shown in the figure below:



Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from decomposition of waste at the SWDS	CO_2	No	CO_2 emissions from the decomposition of fresh waste are not accounted for
		CH_4	Yes	The major source of emissions in the baseline.
		N_2O	No	N_2O emissions are small compared to CH4 emissions from landfills. Exclusion of this gas is conservative
		Other	No	
	Emissions from electricity generation	CO_2	Yes	Major source. Electricity generation emission is included in electricity grid-based emissions.
		CH_4	No	Excluded for simplification. This is conservative.
		N_2O	No	Excluded for simplification. This is conservative.
Project	Emissions from on-site fossil fuel consumption due to the project activity	CO_2	Yes	Includes heat generation for mechanical/thermal treatment process, auxiliary fossil fuels needed to be added into incinerator, etc. It does not include transport.
		CH_4	No	Excluded for simplification. This is conservative.

Source		Gas	Included?	Justification/Explanation
		N_2O	No	Excluded for simplification. This is conservative.
		Other		
	Emissions from on-site electricity use	CO_2	Yes	Electricity consumed for the operation of the project activity will be supplied from the Grid.
		CH_4	No	Excluded for simplification. This is conservative.
		N_2O	No	Excluded for simplification. This is conservative.
	Emission from the waste treatment processes	CO_2	Yes	CO_2 emission generated from combustion of RDF.
		CH_4	No	Excluded for simplification. This is conservative.
		N_2O	No	Excluded for simplification. This is conservative.

3.4 Baseline Scenario

The baseline study was conducted using relevant methodology AMS-III.E (Version 17.0) and AMS-I. D (Version 18.0) as shown below:

As per para. 28 of AMS-III.E, the baseline scenario is the situation where, in the absence of the project activity, organic waste matter is left to decay within the project boundary and methane is emitted to the atmosphere. The yearly baseline emissions are the amount of methane that would have been emitted from the decay of the cumulative quantity of the waste diverted or removed from the disposal site, to date, by the project activity, calculated as the methane generation potential using the Tool04, “Emissions from solid waste disposal sites” Version 08.1

Prior to the implementation of the project activity, MSW in Pimpri-Chinchwad was mainly subjected to landfill in anaerobic condition without any methane recovery system for flaring or energy recovery purpose, therefore in the absence of the project activity, methane emissions from anaerobic decay of waste would have been released into the atmosphere.

As per para. 19 of AMS-I.D, the baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

Prior to the implementation of the project activity, the electricity supply for Pimpri-Chinchwad was generated mainly from grid connected power plants.

3.5 Additionality

The present VCS project generates power using waste to energy which is a renewable, zero emission source of energy. The methodology AMS I D requires the project investor to determine the additionality based on "Methodological Tool 21- Demonstration of additionality of small-scale project activities", Version 13.1.

Power generation using Municipal solid waste is not a legal requirement or a mandatory option. There are state and sectoral policies and schemes, by Ministry of new and renewable energy (MNRE)[24] framed primarily to encourage WTE plants in India. These schemes have also been drafted realizing the extent of risks involved in the projects and to attract private investments. The Solid waste management rule, 2016[25] & Solid waste management rule (2020 Amendment)[26] does not mandate to set up WTE plant for minimizing solid waste in India. There is no legal requirement on the choice of a particular technology for Waste handling and disposal and reduce the methane emission. There is no mandatory requirement to implement the project activity.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☒ No

As per the host country, the following regulatory laws are being applicable for the setting up of Waste to energy business activity in India.

- The Solid waste management rule, 2016
- The Solid waste management rule, 2020 (Amendment)

[24] https://mnre.gov.in/waste-to-energy/

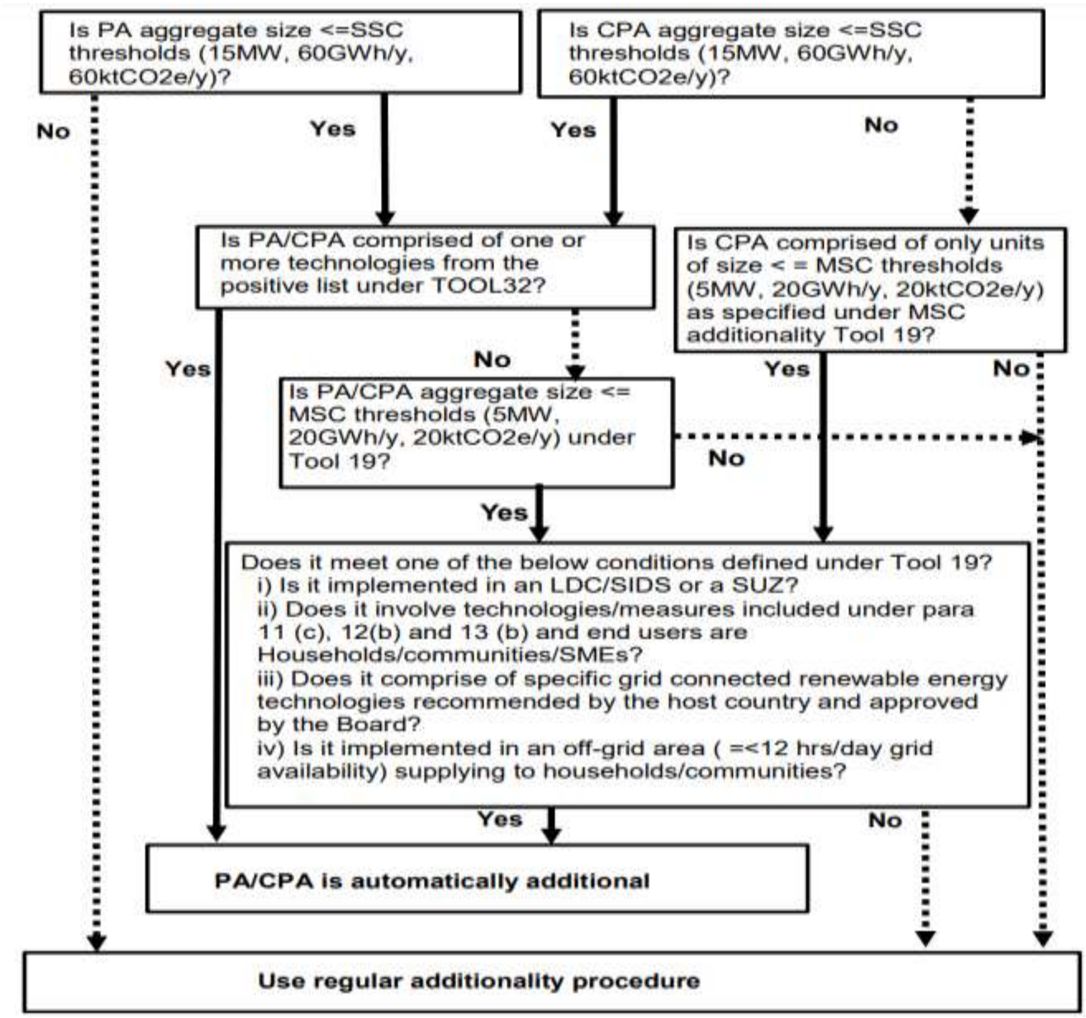
[25] https://cpcb.nic.in/uploads/MSW/MSW_AnnualReport_2001-02.pdf

[26] https://cpcb.nic.in/uploads/MSW/SWM_2016.pdf

There are no legal and regulatory requirements in the host country that mandates for processing of municipal solid waste through waste to energy power plant, this project activity is completely voluntary activity by the project proponent.

3.5.2 Additionality Methods

The step-wise approach to establish additionality of the project activity has been followed, details of which are provided in the following paragraphs:



Step 1: Is PA aggregate size <= SSC thresholds (15 MW, 60 GWh/y, 60 ktCO$_2$e/y)

The size of the project activity is less than 15MW and hence within the SSC thresholds.

Step 2: Is PA/CPA comprised of one or more technologies from the positive list under TOOL 32

This is a waste to energy-based Grid connected Power generation Project. As per the latest version of the Tool 32: Positive List of Technologies" version 04, This technology is not included in the positive list.

Step 3: Is PA/CPA Aggregate size <= MSC thresholds (5 MW, 20 GWh/Y, 20 ktCO_2e/y) under Tool19?

Installed capacity of the project is 14 MW and hence does not come under micro activity thresholds.

Step 4: Does it meet one of the below conditions defined under Tool 19?

1. Is it implemented in an LDC/SIDC or a SUZ?
2. Does it involve technologies/measures included under para 11 (c), 12(b) and 13 (b) and end users are Households/communities/SMEs?
3. Does it comprise of specific grid connected renewable energy technologies recommended by the host country and approved by the Board?
4. Is it implemented in an off-grid area (=<12 hrs/day grid availability) supplying to households/communities?

Justification:

The above said conditionalities does not apply and hence the regular additionality approach needs to be followed as stated below:

As per Methodological Tool 21, Tool for the demonstration of Additionality of Small-scale Project Activities (Version 13.1, EB 105, Annex 4, to establish the project additionality, it has to be shown that the project activity would not have occurred anyway due to at least one of the following barriers:

- **Investment barrier:** a financially more viable alternative to the project activity would have led to higher emissions.
- **Technological barrier:** a less technologically advanced alternative to the project activity involves lower risks due to the performance uncertainty or low market share of the new technology adopted for the project activity and so would have led to higher emissions.
- **Barrier due to prevailing practice:** prevailing practice or existing regulatory or policy requirements would have led to implementation of a technology with higher emissions.
- **Other barriers:** without the project activity, for another specific reason identified by the project proponent, such as institutional barriers or limited information, managerial resources, organizational capacity, financial resources, or capacity to absorb new technologies, emission would have been higher.

The project proponent has selected the Investment barrier to demonstrate in a conservative and transparent manner that the project activity is financially unattractive. In line with the guidelines stipulated under Annex 34 of EB 358 (“Non-binding best practice examples to demonstrate additionality for SSC project activities”), a benchmark analysis has been used for the analysis and Equity IRR has been chosen as the financial indicator for the demonstration of the additionality.

Selection of Appropriate Benchmark:

The Cost of Equity has been considered using the "TOOL27: Investment analysis, version 8.0" available at the time of investment decision dated 1-November-2017 (source) as well as the latest version available. As the project activity generates power utilizing waste to energy, Group 1 as per para 5a of Appendix of EB 122, Annex 1 has been identified as a suitable category.

As per appendix of "TOOL27: Investment analysis, version 8.0", the expected return of equity in real terms for Energy Industries (Group 1) in India = 10.73%.

TOOL27: "Investment Analysis", version 14.0 is the latest version available. The default value mentioned in version 14.0 is 9.13% for group 1 projects in India is used which is appropriate and more conservative for benchmark calculation and the project owner has considered the same tool version for default value of return on equity for the respective project.

The investment analysis has been carried out in Nominal terms. Accordingly, default value as given in table under the Appendix, EB 122, Annex 1 has been adjusted by adding suitable forecasted inflation rate.

The inflation forecast has been considered from the RBI[27] (Reserve Bank Of India) based on the investment decision date (11.01.2018) for 10 years is =4.40%

The benchmark has been computed in the following manner:

Default Value Benchmark:

The cost of equity is determined by selecting the values provided in the table of the Appendix, i.e., Default values for cost of equity (expected return on equity) in the Tool27: Investment analysis, Version 14.0.

The Required return on equity (benchmark) was computed in the following manner:

Nominal Benchmark[28] = {(1+Real Benchmark) * (1+Inflation rate)}-1

Where:

Default value for Real Benchmark = 9.13% (as per Appendix of EB 122, Annex 1)

Inflation Rate forecast by RBI[29].

[27] https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=18046

[28] https://cdm.unfccc.int/Reference/Guidclarif/ssc/methSSC_guid15_v01.pdf

[29] https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=18046

The Inflation rate forecast of the host country; India is published by the RBI (Reserve Bank of India). As per the Tool27: Investment analysis, version 14.0, the average forecasted inflation rate has considered the next 10 years average inflation rate.

Benchmark estimation:

The Cost of Equity has been considered using the “Tool27: Investment analysis” version 14.0 as per the latest available value.

Table under Appendix in EB 122, Annex 1 specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in India= 9.13%

Inflation Forecast	Benchmark
4.40%	13.93%

Hence a benchmark of 13.93% has been selected for the project activity.

Initially, the consortium M/S. Antony Lara Enviro Solutions Pvt Ltd and M/S. AG Enviro Infra Projects Pvt Ltd has participated in the Request for Proposal (RfP) issued by Pimpri Chinchwad Municipal Corporation (PCMC) dated 25-September-2017.

The consortium had submitted the Detailed Project Report (DPR) (prepared by third party) with key assumptions and financial parameters to PCMC along with the bid submission in line with the request for proposal. At this time, the board has started evaluating the input assumptions and financial viability of the project activity to participate in the bidding and further to invest in the project activity and the bid was submitted on 11-January-2018 which is considered as the investment decision date. The letter of award for the implementation of the project activity was received on 28-February-2018. Input values for the investment analysis are sourced from the DPR which is the first and foremost document available at the time when the project proponent started evaluating the project activity for the bidding and investment. Hence, all the input parameters used in the DPR are available at the time of decision making.

Subsequently, in line with the 4.2.3 (f) para of RfP which states “the members of a Consortium shall form an appropriate SPV to execute the Project, if awarded to the Consortium”, the Antony Lara Renewable Energy Private Limited was founded in 24-July-2018 to execute the project.

Chronology of events of the project activity is provided below

S. No	Activity	Date of Activity
1.	Request for proposal issued by Pimpri Chinchwad Municipal Corporation (PCMC)	25-September-2017
2.	Completion of DPR by ESBEE Power Solutions Pvt Ltd.	December 2017
3.	Bid submission to PCMC – **Investment decision date**	11-January-2018
4.	Letter of award issued by PCMC to consortium M/S. Antony Lara Enviro Solutions Pvt Ltd and M/S. AG Enviro Infra Projects Pvt Ltd	28-February-2018
5.	Formation of the SPV for project execution - Antony Lara Renewable Energy Private Limited (ALREPL)	24-July-2018
6.	Signing of Concession Agreement between PCMC and ALREPL	06-September-2018
7.	Signing of Conversion charges Agreement between PCMC and ALREPL	26-June-2020
8.	Signing of EPC Contract between ALREPL and Hitachi Zosen India Private Limited and ISGEC HEAVY Engineering Limited	06-July-2020
9.	Commissioning Date	07-October-2023

Calculation and comparison of financial indicators

The period considered for Post Tax Equity IRR calculations is 21 years, which corresponds to the concession agreement period.

Depreciation, and other non-cash items related to the project activity, which have been deducted in estimating gross profits on which tax is calculated, is added back to net profits for the purpose of calculating the financial indicator.

Appropriateness for consideration of Input Parameters:

1. As per the concession agreement, the tipping fee i.e., revenue from processing of the MSW is 504 Rs/tonne has been paid by PCMC to Antony Lara. This supports the tipping fee provided in the DPR dated December 2017 prepared by ESBEE Power Solutions Pvt. Ltd, a third-party engineering company and the same had been considered for investment analysis.

2. The bottom ash and fly ash has been sent to Construction and Demolition waste processing plant for production of construction materials for free of cost. Hence, the revenue from bottom ash and fly ash has not been considered for the investment analysis.

3. As per para 3 of EB48 Guidelines, Annex 11, the plant load factor shall be defined ex-ante in the CDM-PDD according to one of the following three options,
   a. The plant load factor provided to banks and/or equity financiers while applying the project activity for project financing, or to the government while applying the project activity for implementation approval;
   b. The plant load factor determined by a third party contracted by the project proponent (e.g. an engineering company);

The PLF value considered for investment analysis is 80% which has been sourced from DPR dated December 2017 prepared by ESBEE Power Solutions Pvt. Ltd, a third-party engineering company which is in line with the para 3(b) of "Guidelines for the reporting and validation of plant load factors". Hence, the PLF considered for the investment analysis complies with EB48 Annex 11[30]

4. In India, generally the power purchase agreement (PPA) is signed with fixed tariff rate over the lifetime of the project activity. Hence a fixed tariff rate without escalation has been considered for the investment analysis. Moreover, as per the project activity, the PCMC has signed a conversion charges agreement to utilize the electricity generated by the project through the grid for a fixed tariff of 5 INR/kWh.

5. The O&M cost has been sourced from DPR available at the time of investment decision considering 3% of escalation. The O&M cost is comprised of the spares, employee cost, administrative and general expense, consumables, repairs and maintenance and insurance expenses, etc. Due to the rising costs of labour, materials, equipment and transportation, the inflation of O&M every year has been considered.

Input values considered for the IRR calculations are given below.

Particulars	Value	Unit	Source/Remarks
Quantity of MSW treated	700	TPD	DPR dated December 2017 (Page No. 69)
Capacity of the project	14	MW	DPR dated December 2017 (Page No. 127)
Operating days	330	Days	DPR dated December 2017 (Page No. 127)
PLF	80	%	DPR dated December 2017 (Page No. 127)
Annual Net generation	88.704	GWh/year	Calculated
Auxiliary Electricity Consumption	12.50	%	DPR dated December 2017 (Page No. 127)
Net Electricity Generation	77.62	GWh/year	Calculated
Project cost	2500	INR Million	DPR dated December 2017 (Page No. 127)
Debt	70	%	DPR dated December 2017 (Page No. 127)

[30] https://cdm.unfccc.int/EB/048/eb48_repan11.pdf

Equity	30	%	DPR dated December 2017 (Page No. 127)
Debt	1750.00	INR Million	Calculated
Equity	750.00	INR Million	Calculated
Grant	500.00	INR Million	RFP dated 25-September-2017
Interest rate	10.50	%	DPR dated December 2017 (Page No. 127)
Debt Repayment tenure	14	years	DPR dated December 2017 (Page No. 127)
Moratorium	1	year	DPR dated December 2017 (Page No. 127)
Operation and Maintenance (Spares, employee cost, administrative and general expense, statutory charges, repairs, maintenance and expenses)	210.0	INR Million	DPR dated December 2017 (Page No. 127)
Annual Escalation	3.00	%	DPR dated December 2017 (Page No. 127)
Service tax on O&M services	18.00	%	Heading 9954 or 9983 or 9987
Insurance cost (0.75% of project cost)	18.75	INR Million	DPR dated December 2017 (Page No. 127)
Tariff	5	Rs/kWh	DPR dated December 2017 (Page No. 127)
Tipping Fee	504	Rs/tonne	DPR dated December 2017 (Page No. 127)
Interest on working capital	11.66	%	https://www.cercind.gov.in/2017/orders/05.pdf
Depreciation Rate (Book)	4.76	%	Calculated
Project cost	2500.00	INR Million	DPR dated December 2017 (Page No. 127)
Land cost	0	INR Million	DPR dated December 2017 (Page No. 127)
Gross Depreciable Value	2500.00	INR Million	Calculated
Salvage Value (@ 10%)	250.00	INR Million	Calculated
Net Depreciable Value	2250.00	INR Million	Calculated
Residual value	250.00	INR Million	Calculated
IT Depreciation Rate	7.84	%	http://www.incometaxindia.gov.in/charts%20%20tables/depreciation%20rates.htm
Income tax rate	34.27	%	Calculated from the source given below table (Income tax rate+ Income tax rate*Surcharge) + (Income tax rate+ Income tax rate*Surcharge)*Health & Education cess (%)
MAT rate	21.13	%	Calculated from the source given below table (MAT+ MAT*Surcharge) + (MAT+ MAT*Surcharge)*Health & Education cess (%)
Salvage Value	10	%	DPR dated December 2017 (Page No. 127)
Concession period	21.00	Year	DPR dated December 2017 (Page No. 19)

Financial Year	AY 2018 - 19	Source	
Income tax rate (%)	30.00	Indian IT Act for AY 2018-19	https://incometaxindia.gov.in/_layouts/15/dit/Pages/viewer.aspx?path=/Charts%20%20Tables/Tax-rates-for-last-10-years/Companies.html&IsDlg=0
MAT (%)	18.50	Indian IT Act for AY 2018-20	https://incometaxindia.gov.in/_layouts/15/dit/pages/viewer.aspx?grp=act&cname=CMSID&cval=102120000000073619&searchFilter=&k=&IsDlg=0
Surcharge (%)	12.00	Indian IT Act for AY 2018-21	https://incometaxindia.gov.in/_layouts/15/dit/Pages/viewer.aspx?path=/Charts%20%20Tables/Tax-rates-for-last-10-years/Companies.html&IsDlg=0
Health & Education cess (%)	2.00	Indian IT Act for AY 2018-22	https://incometaxindia.gov.in/_layouts/15/dit/Pages/viewer.aspx?path=/Charts%20%20Tables/Tax-rates-for-last-10-years/Companies.html&IsDlg=0

Post Tax Equity IRR for the project activity against the benchmark values are shown in table below. Thus, it is evident that the project is not financially attractive as the equity IRR is below the benchmark value.

Post tax Equity IRR	
Project post-tax equity IRR	1.64%
Benchmark Value	13.93%

Sensitivity Analysis:

The robustness of the conclusion drawn above, namely that the project is not financially attractive, has been tested by subjecting critical assumptions to reasonable variation. As required by EB 122, Annex 1, only variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenues should be subjected to reasonable variation. PP has identified the total revenue from the project activity is dependent on the Tariff, Plant Load Factor, Project Cost, and O&M Costs and debt constitute more than 20% of the project costs. These factors have been subjected to a 10% variation on either side and the results of the sensitivity analysis indicate that even after applying such variation the EIRR does not cross the benchmark.

Variation %	-10%	Normal	10%	Variation required to reach benchmark	Value required to reach benchmark
Tariff (Rs/KWh)	-1.71%	1.64%	4.84%	39.14%	6.957
PLF (%)	-1.71%	1.64%	4.84%	39.14%	111.31%
Project Cost (INR Million)	3.73%	1.64%	-0.09%	-40.10%	1497.50
O&M Cost (INRV Million)	4.41%	1.64%	-1.40%	-49.40%	106.26

Debt (INR Million)	3.29%	1.64%	0.14%	-61.20%	679

a) **Tariff:** The Tariff rate of electricity used for investment analysis i.e., 5.00 INR/kWh is sourced from DPR. The post-tax equity IRR will breach the benchmark value at a tariff variation of 39.14%. However, the actual tariff based on the concession agreement signed is same as the estimated tariff. Hence, increase in tariff is unlikely and the benchmark will not be breached.

b) **PLF:** The PLF value considered for investment analysis is 80% sourced from DPR and complies with EB48 Annex 11[31] as the DPR was prepared by third party engineering company. The post-tax equity IRR breaches the benchmark value at a PLF variation of 39.14%. The increase in PLF value to breach the benchmark is highly unlikely as the normative PLF of the MSW project is less than the estimated PLF.

c) **Project Cost:** The project cost considered for investment analysis i.e., 2500 million INR which is sourced from DPR. A variation of -40.10% is required for post-tax equity IRR to breach benchmark which is not possible as the project is already commissioned. The actual cost incurred in commissioning of the project is 2338.73 million INR at which the post-tax equity IRR has not breached the benchmark. Hence, reduction in 40.10% is highly unlikely to happen.

d) **O&M Costs:** The Operation and Maintenance cost considered for investment analysis is 210 Mn INR sourced from DPR. A variation of -49.40% is required for post-tax equity IRR to breach benchmark which is not possible. The post-tax equity IRR has not breached the benchmark with the actual O&M cost i.e., 330 Mn INR. Hence, the reduction of 49.40% is unlikely to happen.

An analysis has been done to identify the percentage variation at which the financial indicators will equal/breach the benchmark and the probability of its occurrence. Based on sensitivity analysis it can be concluded that the project activity is additional with reasonable variation in values and is not likely to breach the benchmark value. The occurrence of these events is unlikely.

Conclusion:

As described above, the project fulfils all necessary requirements of additionality specified in the Tool 21- Demonstration of Additionality of Small-scale Project Activities, version 13.1. Hence, the project is deemed additional.

[31] https://cdm.unfccc.int/EB/048/eb48_repan11.pdf

3.6 Methodology Deviations

There are no deviations in the applied methodology.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Baseline Emissions from Solid Waste Disposal Sites

As per para 17 of Tool 04: "Emissions from solid waste disposal sites", version 08.1, the baseline emissions for a year are calculated using the following formula:

$$\left.\begin{array}{l} BE_{CH4,SWDS,y} \\ PE_{CH4,SWDS,y} \\ LE_{CH4,SWDS,y} \end{array}\right\} = \varphi_y \times (1 - f_y) \times GWP_{CH4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y}$$
$$\times MCF_y \times \sum_{x=1}^{y} \sum_{j} \left(W_{j,x} \times DOC_j \times e^{-k_j \times (y-x)} \times (1 - e^{-k_j}) \right)$$

Equation (1)

Where,

$BE_{CH4,SWDS,y}$ $PE_{CH4,SWDS,y}$ $LE_{CH4,SWDS,y}$	= Baseline, project or leakage methane emissions occurring in year y generated from waste disposal at a SWDS during a time period ending in year y (t CO_2e/yr)
X	= Years in the time period in which waste is disposed at the SWDS, extending from the first year in the time period (x = 1) to year y (x = y)
y	= Year of the crediting period for which methane emissions are calculated (y is a consecutive period of 12 months)
$DOC_{f,y}$	= Fraction of degradable organic carbon (DOC) that decomposes under the specific conditions occurring in the SWDS for year y (weight fraction)
$W_{j,x}$	= Amount of solid waste type j disposed or prevented from disposal in the SWDS in the year x (t)
φ_y	= Model correction factor to account for model uncertainties for year y

f_y = Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y

GWP_{CH4} = Global Warming Potential of methane

OX = Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)

F = Fraction of methane in the SWDS gas (volume fraction)

MCF_y = Methane correction factor for year y

DOC_j = Fraction of degradable organic carbon in the waste type j (weight fraction)

k = Decay rate for the waste type j (1 / yr)

j = Type of residual waste or types of waste in the MSW

As per para 27, equation (5) of tool 04, version 08.1, the amount of solid waste type j disposed or prevented from disposal in the SWDS in the year x is calculated as follows:

$$W_{j,x} = W_x \times p_{j,x}$$

Where,

$W_{j,x}$ = Amount of solid waste type j disposed or prevented from disposal in the SWDS in the year x (t)

W_x = Total amount of solid waste disposed or prevented from disposal in the SWDS in year x (t)

$p_{j,x}$ = Average fraction of the waste type j in the waste in year x (weight fraction)

j = Type of residual waste or types of waste in the MSW

x = Years in the time period for which waste is disposed at the SWDS, extending from the first year in the time period (x = 1) to year y (x = y)

As per para 28, equation (7) of tool 04, version 08.1, the fraction of the waste type j in the waste for the year x or month i are calculated as follows:

$$p_{j,x} = \frac{\sum_{n=1}^{z_x} p_{n,j,x}}{Z_x}$$

Where,

$p_{j,x}$ = Average fraction of the waste type j in the waste in year x (weight fraction)

Z_x = Number of samples collected during the year x

$p_{n,j,x}$ = Fraction of the waste type j in the sample n collected during the year x (weight fraction)

n = Samples collected in year x

The estimation of baseline emissions from Solid Waste Disposal Sites as follows:

φ_y	Model Correction Factor	0.80

1-fy	Fraction of methane captured	1.0
GWP_{CH4}	Global Warming Potential of CH_4	28
1-OX	Oxidation factor	0.9
F	Fraction of CH_4	0.5
$DOC_{f,y}$	Fraction of DOC decomposed	0.5
MCF_y	Methane Correction Factor	1

Quantity of RDF processed per day = 700 TPD
Operating days in year x = 330 days
Operating hours in day = 24 hours
Quantity of RDF processed per year (W_x) = 231,000 Tonnes/year

$$\boldsymbol{W_{j,x} = W_x \times p_{j,x}}$$

Composition	**$P_{j,x}$**	**W_{jx}**	**DOC_j**	**K_j**	**W_{jx}**
j	%	TPY (Wet)	%	1/y	TPD (Wet)
Food	0.00%	0	15%	0.085	0.00
Paper	19.71%	45,530	40%	0.045	137.97
Wood including Coconut Shell	2.20%	5,082	43%	0.025	15.40
Green/Mix	46.39%	107,161	20%	0.065	324.73
Plastic	8.70%	20,097	0%	0.000	60.90
Cloth	20.59%	47,563	24%	0.045	144.13
Rubber	0.00%	0	0%	0.045	0.00
Metal	0.00%	0	0%	0	0.00
Glass	0.70%	1,617	0%	0	4.90
Inert	1.71%	3,950	0%	0	11.97
Total	**100.00%**	**231,000**			**700**

Year	Emission from Food	Emission from paper	Emission from wood including coconut shell	Emission from green mix	Emission from cloth	$BE_{CH4,SWDS,y}$ (tCO_2e/ year)
1	0	5,721.8	385.2	9630.3	3586.4	**19,324**
2	0	11,191.9	761.0	18,654.5	7014.9	**37,622**
3	0	16,421.2	1127.4	27,110.9	10,292.6	**54,952**
4	0	21,420.5	1484.8	35,035.0	13,426.1	**71,366**
5	0	26,199.7	1833.4	42,460.5	16,421.7	**86,915**

6	0	30,768.7	2173.3	49,418.6	19,285.5	**101,646**
7	0	35,136.6	2504.9	55,938.9	22,023.2	**115,604**
Total Baseline Emissions from RDF combustion						**487,430**

Baseline Emissions from Electricity Generation

As per para 22 of AMS - I.D," Grid connected renewable electricity generation" version 18.0, the baseline emissions include only CO_2 emissions from electricity generation in power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated using equation (1) of AMS-I. D, version 18.0 as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y} \quad \text{Equation (1)}$$

Where,

BE_y = Baseline emissions in year y (t CO_2)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

$EF_{grid,y}$ = Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO_2/MWh)

The estimation of baseline emissions from electricity generation is as follows:

Details	**Units**	**Values**
Power generation capacity	MW	14
Plant Load Factor	%	80
Operating days	days	330
Annual power generation	MWh/year	88,704
Auxiliary Electricity Consumption	%	12.5
Annual Net power generation	MWh/year	77,616

Combined margin emissions factor

As per para 14 of Tool07,"Tool to calculate the emission factor for an electricity system "version 07.0 the following steps have been followed.

(a) **Step 1:** Identify the relevant electricity systems;
(b) **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional);
(c) **Step 3:** Select a method to determine the operating margin (OM);
(d) **Step 4:** Calculate the operating margin emission factor according to the selected method;

(e) **Step 5:** Calculate the build margin (BM) emission factor;
(f) **Step 6:** Calculate the combined margin (CM) emission factor.

Step 1: Identify the relevant electricity systems

As described in tool07 "For determining the electricity emission factors, identify the relevant project electricity system. Similarly, identify any connected electricity systems". It also states that "If the DNA of the host country has published a delineation of the project electricity system and connected electricity systems, these delineations should be used". Keeping this into consideration, the Central Electricity Authority (CEA), Government of India has divided the Indian Power Sector into five regional grids viz. Northern, Eastern, Western, North-eastern and Southern. However, since August 2006, all regional grids except the Southern Grid had been integrated and were operating in synchronous mode, i.e., at same frequency. Consequently, the Northern, Eastern, Western and North-Eastern grids were treated as a single grid named as NEWNE grid from FY 2007-08 onwards for the purpose of this CO_2 Baseline Database. As of 31 December 2013, the Southern grid has also been synchronized with the NEWNE grid; hence forming one unified Indian Grid. Since the project supplies electricity to the Indian grid, emissions generated due to the electricity generated by the Indian grid as per CM calculations will serve as the baseline for this project.

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional)
Project participants may choose between the following two options to calculate the operating margin and build margin emission factor:

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.
The project participants has chosen only grid power plants in the calculation.

Step 3: Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods, which are described under Step 4:
(a) Simple OM; or
(b) Simple adjusted OM; or
(c) Dispatch data analysis OM; or
(d) Average OM.
The data required to calculate Simple adjusted OM and Dispatch data analysis OM is not possible due to lack of availability of data to project developers. The choice of other two options for calculating operating margin emission factor depends on generation of electricity from low-cost/must-run sources. In the context of the methodology low cost/must run resources typically include hydro, geothermal, Solar, low-cost biomass, nuclear and solar generation.

Share of Must Run (Hydro/Nuclear) (% of Net generation)					
India	2018-19	2019-20	2020-21	2021-22	2022-23
	14.5%	17.0%	16.5%	15.8%	15.33%

Source: Central Electricity Authority (CEA) Database Version 19, 2023[32]

The above data clearly shows that the percentage of total grid generation by low-cost/ must-run plants (on the basis of average of five most recent years) for the Indian grid is less than 50 % of the total generation. Thus, the Average OM method cannot be applied, as low cost/must run resources constitute less than 50% of total grid generation.

The simple OM emission factor is calculated as the generation-weighted average CO_2 emissions per unit net electricity generation (tCO_2/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units.

For the simple OM, the simple adjusted OM and the average OM, the emissions factor can be calculated using either of the two following data vintages:

(a) **Ex-ante option:** if the ex-ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required.

OR

(b) **Ex-post option:** if the ex-post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring.

PP has chosen ex-ante option for calculation of Simple OM emission factor using a 3-year generation-weighted average, based on the most recent data available at the time of submission of the PD to the VCS verifier for verification. OM determined at validation stage will be the same throughout the crediting period. There will be no requirement to monitor & recalculate the emission factor during the crediting period.

Step 4: Calculate the operating margin emission factor ($EF_{grid,OM\ Simple,y}$) according to the selected method:

The operating margin emission factor has been calculated using a 3-year data vintage:

Net Generation in Operating Margin (GWh) (incl. Imports)			
INDIAN Grid	2020-21	2021-22	2022-23

32 https://cea.nic.in/wp-content/uploads/baseline/2024/04/CO2_DatabaseVersion_19_2022_23.xlsx

	958,218	1,035,672	1,117,846
Simple Operating Margin (tCO_2/MWh) (incl. Imports)			
INDIAN Grid	2019-20	2020-21	2022-23
	0.9402	0.9605	0.9710

Weighted Generation Operating Margin	
INDIAN Grid	0.9580

STEP 5: Calculate the build margin emission factor ($EF_{BM,y}$):

Option 1 as described above is chosen to calculate the build margin emission factor for the project activity. BM is calculated ex-ante based on the most recent information available at the time of submission of PDD and is fixed for the entire crediting period.

Build Margin (tCO_2/MWh) (not adjusted for imports)	
INDIAN Grid	2022-2023
	0.8670

STEP 6: Calculate the combined margin (CM) emissions factor:

Combined Margin – The combined margin is the weighted average of the simple operating Margin and the build margin. In particular, for intermittent and non-dispatch able generation types such as wind and solar photovoltaic, Tool07: Tool to calculate the emission factor for an electricity system, Version 07.0.0, EB 100, Annex 100, Annex 4, allows to weigh the operating margin and Build margin at 75% and 25%, respectively for wind and solar projects and 50% and 50%, respectively for hydro and biomass projects.

As per para 85 of Tool 07: Tool to calculate the emission factor for an electricity system, Version 07.0, the combined margin emissions factor is calculated as follows:

$$\mathrm{EF_{grid,CM,y}} = \mathrm{EF_{grid,OM,y}} \times \mathrm{w_{OM}} + \mathrm{EF_{grid,BM,y}} \times \mathrm{w_{BM}}$$

Where,

$EF_{grid,BM,y}$= Build margin CO_2 emission factor in year y (t CO_2/MWh)

$EF_{grid,OM,y}$= Operating margin CO_2 emission factor in year y (t CO_2/MWh)

w_{OM}= Weighting of operating margin emissions factor (per cent)

w_{BM}= Weighting of build margin emissions factor (per cent)

Combined Margin Emission Factor ($EF_{grid,CM,y}$) = 0.50*0.9580+ 0.50*0.8670 = 0.9125 tCO_2/MWh

Total power generation = 14*80%*330*24
= 88,704 MWh/year

Auxiliary Power Consumption = 12.5%

Net Power Generation = 88,704 *(1-0.125)

= 77,616 MWh/year

Total Baseline Emission reduction = 77,616 * 0.9125

= 70,824 tCO_2e/year

Total Baseline Emission reduction from electricity generation = 70,824 tCO_2e/year

Total baseline emissions from waste disposal and electricity generation = 57,719+70,824

= 128,543 tCO_2e/year

4.2 Project Emissions

As per para 22 of AMS- III. E," Avoidance of methane production from decay of biomass through controlled combustion, gasification or mechanical/thermal treatment ", version 17.0, the project emissions consist of

$$\mathrm{PE}_{\mathrm{y}} = \mathrm{PE}_{\mathrm{y,comb}} + \mathrm{PE}_{\mathrm{y,transp}} + \mathrm{PE}_{\mathrm{y,power}}$$

Where,

PE_{y} = Project activity direct emissions in the year y (t CO_2e)

$PE_{y,comb}$ = Emissions through combustion and gasification of non-biomass carbon of waste and RDF/SB in the year y (t CO_2e)

$PE_{y,transp}$ = Emissions through incremental transportation in the year y (t CO_2e)

$PE_{y,power}$ = Emissions through electricity or diesel consumption in the year y (t CO_2e)

Project emissions from combustion process of RDF:

As per para 24 of AMS- III.E, "Avoidance of methane production from decay of biomass through controlled combustion, gasification or mechanical/thermal treatment", version 17.0, the project emissions from combustion of combustion of the non-biomass (i.e., fossil) carbon content of the wastes and RDF/SB and from the auxiliary fossil fuel consumed will be estimated as follows:

$$PE_{y,comb} = Q_{y,non-biomass} \times 44/12 + Q_{y,fuel} + EF_{y,fuel}$$

Where,

$Q_{y,non-biomass}$ = Non-biomass carbon of the waste and RDF/SB combusted/gasified in the year y (tonnes of carbon)

$Q_{y,fuel}$ = Quantity of auxiliary fossil fuel used in the year y (tonnes)

$EF_{y,fuel}$ = CO_2 emission factor for the combustion of the auxiliary fossil fuel (tonnes CO_2 per tonne fuel, according to latest IPCC Guidelines)

The estimation of project emission from combustion of RDF is as follows:

$$Q_{non-biomass} = W_{j,x} \times Default\ dry\ matter\ content\ (\%) \times FCC_{j,y} \times FFC_{j,y}$$

Where,

$W_{j,x}$ = Quantity of waste type j fed into combustor c the in year x (dry basis)

$FCC_{j,y}$ = Fraction of total carbon content in waste type j in year x

$FFC_{j,y}$ = Fraction of fossil carbon in total carbon content of waste type j in year x (weight fraction)

Description	W_{jx} (MT)	Default wet matter content	Default dry matter content	W_{jx}	$FCC_{j,y}$	$FFC_{j,y}$	$Q_{y,non,biomass}$ (MT)
j	**TPY (Wet)**	(%)	(%)	TPY (Dry)	Total Carbon (%)	Fossil carbon in total carbon (%)	**Non-biomass carbon of the RDF (TPY)**
Food	0	60%	40%	0	38%	0%	0
Paper	45,530	10%	90%	40,977	46%	1%	188
Wood including Coconut Shell	5,082	15%	85%	4,320	50%	0%	0
Green/Mix	107,161	60%	40%	42,864	49%	0%	0
Plastic	20,097	0%	100%	20,097	75%	100%	15,073
Cloth	47,563	20%	80%	38,050	50%	20%	3,805
Rubber	0	16%	84%	0	67%	20%	0
Metal	0	0%	100%	0	0%	0%	0
Glass	1,617	0%	100%	1,617	0%	0%	0
Inert	3,950	10%	90%	3,555	3%	100%	107
Total	**231,000**			**151,481**			**19,173**

Parameter	Description	Unit	Value
$Q_{ynon,biomass}$	Non-biomass carbon of the waste and RDF/SB combusted/gasified in the year y (tonnes of carbon)	tonnes	19,173

Q_{yfuel}	Quantity of auxiliary fossil fuel used in the year y	litres	29,232
$\rho_{i,y}$	Density of Diesel	kg/litre	0.88
Q_{yfuel}	Quantity of auxiliary fossil fuel used in the year y	tonnes	25.724
EF_{yfuel}	Weighted average CO_2 emission factor of fuel (diesel)	Kg CO_2/TJ	74,800
$NCV_{i,y}$	Net calorific value of fuel (diesel)	TJ/Gg	43.3
EF_{yfuel}	CO2 emission factor for the combustion of the auxiliary fossil fuel (tonnes CO_2 per tonne fuel, according to latest IPCC Guidelines)		3.24

$$PE_{y,comb} = Q_{y,non-biomass} \times \frac{44}{12} + Q_{y,fuel} \times EF_{y,fuel}$$

Total project emissions from combustion of RDF ($PE_{y,comb}$) = (19,173 * 44/12) + (25.724*3.24)
=70,384 tCO_2e/year

Project Emissions from fossil fuel combustion

As per para 6, equation (1) of tool 03, version 03.0, the project emissions through diesel consumption are determined as follows:

$$PE_{FC,j,y} = \sum_i FC_{i,j,y} \times COEF_{i,y}$$

Where,

$PE_{FC,j,y}$ = CO_2 emissions from fossil fuel combustion in process j during the year y (tCO_2/yr)

$FC_{i,j,y}$ = Quantity of fuel type i combusted in process j during the year y (mass or volume unit/yr)

$COEF_{i,y}$ = CO_2 emission coefficient of fuel type i in year y (tCO_2/mass or volume unit)

As per para 7 (b), equation (4) of tool 03, version 03.0, the CO_2 emission coefficient $COEF_{i,y}$ is calculated based on net calorific value and CO_2 emission factor of the fuel type i, as follows:

$$COEF_{i,y} = NCV_{i,y} \times EF_{CO2,i,y}$$

Where,

$NCV_{i,y}$ = Weighted average net calorific value of the fuel type i in year y (GJ/mass or volume unit)

$FC_{i,j,y}$	Quantity of fossil fuel combusted in the project activity (diesel)	Litres	4080
$\rho_{i,y}$	Density of Diesel	kg/litre	0.88
$FC_{i,j,y}$	Quantity of fossil fuel combusted in the project activity	tonnes	3.6
$NCV_{i,y}$	Net calorific value of fuel (diesel)	TJ / Gg	43.3
$EF_{CO2,i,y}$	Weighted average CO_2 emission factor of fuel (diesel)	Kg/TJ	74800
$COEF_{i,y}$	CO_2 emission factor for the combustion of the fossil fuel		3.24

$EF_{CO2,i,y}$ = Weighted average CO_2 emission factor of fuel type i in year y (tCO_2/GJ)

The estimation of project emission from fossil-fuel combustion is as follows:

Project Emissions from fossil fuel combustion = (3.6 *3.24)

$PE_{FC,j,y}$ = 11 tCO_2e/year

Project Emissions from Incremental Transportation

$$PE_{TR,m} = \sum_f D_{f,m} \times FR_{f,m} \times EF_{CO2,f} \times 10^{-6}$$

Where,

$PE_{TR,m}$ = Project emissions from transportation of freight monitoring period m (t CO_2)

$D_{f,m}$ = Return trip distance between the origin and destination of freight transportation activity f in monitoring period m (km)

$FR_{f,m}$ = Total mass of freight transported in freight transportation activity f in monitoring period m (t)

$EF_{CO2,f}$ = Default CO2 emission factor for freight transportation activity f (g CO_2/t km) *f* = *Freight transportation activities conducted in the project activity in monitoring period m*

Parameter	Description	Unit	Value	Remark
$D_{MSW,m}$	Return trip distance between the origin and destination of freight transportation for collection of MSW	km	0	The collection and transportation of MSW is under the scope of Pimpri-Chinchwad Municipal Corporation. The average distance between the project site and waste collection point is in same range of distance between the waste collection point and the landfill site in the baseline scenario. Hence the project emissions from freight transportation for collection of MSW is considered as zero.
$FR_{MSW,m}$	Total mass of freight transported in freight transportation for collection of MSW	t	700	
N	No of Operating days	days	330	
$D_{Inert,m}$	Return trip distance between the origin and destination of freight transportation for disposal of inert waste	km	0	The inert waste has been disposed in the Landfill which is in project site and hence, the return trip distance between the origin and destination of freight transportation for disposal of inert waste is considered as zero
$FR_{inert,m}$	Total mass of freight transported in freight transportation for disposal of inert waste	TPD	142.85	Plant Record
$D_{ash,m}$	Return trip distance between the origin and destination of freight transportation for disposal of combustion residue	km	0	The fly ash and bottom ash has been sent to Construction & Demolition waste processing plant. Hence, the return trip distance between the origin and destination of freight transportation for disposal of combustion residue is zero.
$F_{ash,m}$	Total mass of freight transported in freight transportation for disposal of combustion residue	TPD	77.38	Plant Record
$EF_{CO2,f}$	Default CO_2 emission factor for freight transportation activity f	g CO_2/t km	245	As per table 1 of tool 12, version 01.1.0
$PE_{y,transp}$	**Project Emissions from Incremental Transportation**	**tCO_2/ year**	**0**	Calculated

Total Project emissions = $PE_{y,comb}$ + $PE_{FC,j,y}$ + $PE_{y,transp}$

= 70,384 + 11 + 0

PE_y = 70,395 tCO_2/annum

4.3 Leakage Emissions

According to the applied methodology of AMS- III.E, "Avoidance of methane production from decay of biomass through controlled combustion, gasification or mechanical/thermal treatment"*version* 17.0, para 37, the RDF produced is not subjected to anaerobic conditions before its combustion. Hence, no leakage emission is considered for incineration of RDF.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

As per para 43, equation 9 of AMS.I.D, "Grid connected renewable electricity generation" version. 18.0, Emission reductions are calculated as follows

$$ER_y = BE_y - PE_y - LE_y$$

ER_y = Emission reductions in year y (t CO_2e)

BE_y = Baseline emissions in year y (t CO_2e)

PE_y = Project emissions in year y (t CO_2e)

LE_y = Leakage emissions in year y (t CO_2e)

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated buffer pool allocation (tCO_2e)	Estimated reduction VCUs (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCU issuance (tCO_2e)
07 - October-2023 to 31-December-2023	20,961	16,586	0	0	4,374	0	4,374
01 - January-2024 to 31-December-2024	93,081	70,395	0	0	22,685	0	22,685

01 - January-2025 to 31-December-2025	110,076	70,395	0	0	39,681	0	39,681
01 - January-2026 to 31-December-2026	126,184	70,395	0	0	55,788	0	55,788
01 - January-2027 to 31-December-2027	141,388	70,395	0	0	70,992	0	70,992
01 - January-2028 to 31-December-2028	155,946	70,395	0	0	85,550	0	85,550
01 - January-2029 to 31-December-2029	169,586	70,395	0	0	99,191	0	99,191
01 - January-2030 to 06-October-2030	137,304	53,809	0	0	83,495	0	83,495
Total	954,525	492,765	**0**	**0**	**461,756**	**0**	**461,756**
Annual Average	**119,315**	**61,595**	**0**	**0**	**57,719**	0	**57,719**

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	GWP_{CH4}
Data unit	tCO_2e/tCH_4
Description	Global Warming Potential for methane
Source of data	2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories, 5th Assessment Report[33]
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	For the calculation of the Baseline Emission
Comments	NA

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO_2e/MWh
Description	Operating Margin CO_2 emission factor in year y
Source of data	CO_2 Emission Database, Version 19.0, Nov- 2023 published by Central Electricity Authority (CEA), Government of India.[34]
Value applied	0.9580
Justification of choice of data or description of measurement methods and procedures applied	NA
Purpose of Data	For the calculation of the Baseline Emission
Comments	NA

Data / Parameter	$EF_{grid,BM,y}$

[33] https://ghgprotocol.org/sites/default/files/ghgp/Global-Warming-Potential-Values%20%28Feb%2016%202016%29_1.pdf

[34] https://cea.nic.in/wp-content/uploads/baseline/2024/04/CO2_DatabaseVersion_19_2022_23.xlsx

Data unit	tCO_2e/MWh
Description	Build Margin CO_2 emission factor in year y
Source of data	CO_2 Emission Database, Version 19.0, Nov- 2023 published by Central Electricity Authority (CEA), Government of India.[35]
Value applied	0.8670
Justification of choice of data or description of measurement methods and procedures applied	NA
Purpose of Data	For the calculation of the Baseline Emission
Comments	NA

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO_2e/MWh
Description	Combined Margin CO_2 emission factor in year y
Source of data	CO_2 Emission Database, Version 19.0, Nov- 2023 published by Central Electricity Authority (CEA), Government of India.[36]
Value applied	0.9125
Justification of choice of data or description of measurement methods and procedures applied	The combined margin emissions factor is calculated as follows: $EF_{grid,CM,y} = EF_{grid,OM,y} * W_{OM} + EF_{grid,BM,y} * W_{BM}$ Where, $EF_{grid,OM,y}$ = Operating margin CO_2 emission factor in year y (tCO_2/MWh) $EF_{grid,BM,y}$ = Build margin CO_2 emission factor in year y (tCO_2/MWh) W_{OM} = Weighting of operating margin emissions factor (%) = 50% W_{BM} = Weighting of operating margin emissions factor (%) = 50%
Purpose of Data	For the calculation of baseline emissions
Comments	NA

[35] https://cea.nic.in/wp-content/uploads/baseline/2024/04/CO2_DatabaseVersion_19_2022_23.xlsx

[36] https://cea.nic.in/wp-content/uploads/baseline/2024/04/CO2_DatabaseVersion_19_2022_23.xlsx

Data / Parameter	$\emptyset_{default}$
Data unit	-
Description	Default value for the model correction factor to account for model uncertainties
Source of data	Default
Value applied	0.80
Justification of choice of data or description of measurement methods and procedures applied	For baseline emissions: refer to the table below to identify the appropriate factor based on the application of the tool (A or B) and the climate where the SWDS is located. Default values for the model correction factor (see table below) The project site falls under tropical climatic condition, where the mean annual temperature is greater than 20°C and the mean annual precipitation is less than 1000 mm. Hence the default value for the model correction factor to account for model uncertainties for application B is 0.80.
Purpose of Data	For the calculation of the Baseline Emission
Comments	NA

Default values for the model correction factor

	Humid/wet conditions	**Dry conditions**
Application A	0.75	0.75
Application B	0.85	0.80

Data / Parameter	F
Data unit	-
Description	Fraction of methane in the SWDS gas (volume fraction)
Source of data	Default value from table 3 of tool 04, version 08.1 as per IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	0.5

Justification of choice of data or description of measurement methods and procedures applied	IPCC 2006 Guidelines for National Greenhouse Gas Inventories[37]
Purpose of Data	For the calculation of the Baseline Emission
Comments	NA

Data / Parameter	OX
Data unit	-
Description	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
Source of data	Default value from table 2 of tool 04, version 08.1
Value applied	0.1
Justification of choice of data or description of measurement methods and procedures applied	Based on an extensive review of published literature on this subject, including the IPCC 2006 Guidelines for National Greenhouse Gas Inventories[38]
Purpose of Data	For the calculation of the Baseline Emission
Comments	NA

Data / Parameter	$DOC_{f,default}$
Data unit	Weight fraction
Description	Default value for the fraction of degradable organic carbon (DOC) in MSW that decomposes in the SWDS
Source of data	Default value from table 4 of tool 04, version 08.1 as per IPCC 2006 Guidelines for National Greenhouse Gas Inventories[39]

[37] https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol5.html

[38]

[39] https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol5.html

Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Purpose of Data	For the calculation of the Baseline Emission
Comments	NA

Data / Parameter	$MCF_{default}$
Data unit	-
Description	Methane correction factor
Source of data	Default value from table 5 of tool 04, version 08.1 as per IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	1.0
Justification of choice of data or description of measurement methods and procedures applied	In case that the SWDS does not have a water table above the bottom of the SWDS and in case of application A, then select the applicable value from the following: (a) 1.0 for **anaerobic managed solid waste disposal sites.** These must have controlled placement of waste (i.e., Waste directed to specific deposition areas, a degree of control of scavenging and a degree of control of fires) and will include at least one of the following: (i) cover material; (ii) mechanical compacting; or (iii) levelling of the waste; (b) 0.5 for **semi-aerobic managed solid waste disposal sites.** These must have controlled placement of waste and will include all of the following structures for introducing air to the waste layers: (i) permeable cover material; (ii) leachate drainage system; (iii) regulating pondage; and (iv) gas ventilation system; (c) 0.8 for **unmanaged solid waste disposal sites – deep.** This comprises all SWDS not meeting the criteria of managed SWDS and which have depths of greater than or equal to 5 meters; (d) 0.4 for **unmanaged-shallow solid waste disposal sites or stockpiles that are considered SWDS.** This comprises all SWDS not meeting the criteria of managed SWDS and which have depths of less than five meters. This includes stockpiles of solid waste that are considered SWDS (according to the definition given for a SWDS)

Purpose of Data	For the calculation of the Baseline Emission
Comments	NA

Data / Parameter	DOC_j
Data unit	Weight fraction
Description	Fraction of degradable organic carbon in the waste type j (weight fraction)
Source of data	Default value from table 6 of tool 04, version 08.1 as per IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Tables 2.4 and 2.5)[40]
Value applied	For MSW, the following values for the different waste types j should be applied: Default values for DOCj **Waste type j** — **DOCj (% wet waste)** Wood and wood products — 43 Pulp, paper and cardboard (other than sludge) — 40 Food, food waste, beverages and tobacco (other than sludge) — 15 Textiles — 24 Garden, yard and park waste — 20 Glass, plastic, metal, other inert waste — 0
Justification of choice of data or description of measurement methods and procedures applied	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Tables 2.4 and 2.5)
Purpose of Data	For the calculation of the Baseline Emission
Comments	NA

Data / Parameter	K_j
Data unit	1/yr

[40] https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/5_Volume5/V5_2_Ch2_Waste_Data.pdf

Description	Decay rate for the waste type j
Source of data	Default value from table 6 of tool 04, version 08.1 as per IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Table 3.3)
Value applied	For MSW, the following values for the different waste types j should be applied: Default values for Kj **Waste type j** / k_j: Pulp, paper, cardboard (other than sludge), textiles: 0.045 Wood, wood products and straw: 0.025 Other (non-food) organic putrescible garden and park waste: 0.065 Food, food waste, sewage sludge, beverages and tobacco: 0.085
Justification of choice of data or description of measurement methods and procedures applied	The project site falls under tropical climatic condition, where the mean annual temperature is greater than 20°C and the mean annual precipitation is less than 1000 mm[41]. Hence the default value for the decay rate for waste type is taken from table 7 of tool 04, version 08.1.
Purpose of Data	For the calculation of the Baseline Emission
Comments	NA

Default values for Kj (table within "Value applied"):

Waste type j	k_j
Pulp, paper, cardboard (other than sludge), textiles	0.045
Wood, wood products and straw	0.025
Other (non-food) organic putrescible garden and park waste	0.065
Food, food waste, sewage sludge, beverages and tobacco	0.085

Data / Parameter	FFC_j
Data unit	%
Description	Fraction of fossil carbon in total carbon content of waste type j
Source of data	Table 2.4, chapter 2, volume 5 of IPCC 2006 guidelines

[41] https://pune.gov.in/about-pune/district-at-a-glance/

Value applied	For MSW the following values for the different waste types j may be applied: Default values for FFC_j <table><tr><th>Waste type j</th><th>Fossil carbon fraction in % of total carbon</th></tr><tr><td>Paper/cardboard</td><td>1</td></tr><tr><td>Textiles</td><td>20</td></tr><tr><td>Food waste</td><td>-</td></tr><tr><td>Wood</td><td>-</td></tr><tr><td>Garden and Park waste</td><td>0</td></tr><tr><td>Nappies</td><td>10</td></tr><tr><td>Rubber and Leather</td><td>20</td></tr><tr><td>Plastics</td><td>100</td></tr><tr><td>Metal</td><td>NA</td></tr><tr><td>Glass</td><td>NA</td></tr><tr><td>Other, inert waste</td><td>100</td></tr></table>
Justification of choice of data or description of measurement methods and procedures applied	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Table 2.4)
Purpose of Data	For the calculation of the Project Emission
Comments	NA

Data / Parameter	FCC_j
Data unit	%
Description	Fraction of total carbon content in waste type j
Source of data	Table 2.4, chapter 2, volume 5 of IPCC 2006 guidelines[42]
Value applied	For MSW the following values for the different waste types j may be applied: Default values for FCC_j <table><tr><th>Waste type j</th><th>Total carbon content in % of dry weight</th></tr><tr><td>Paper/cardboard</td><td>46</td></tr></table>

[42] https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/5_Volume5/V5_2_Ch2_Waste_Data.pdf

	Textiles	50
	Food waste	38
	Wood	50
	Garden and Park waste	49
	Nappies	70
	Rubber and Leather	67
	Plastics	75
	Metal	NA
	Glass	NA
	Other, inert waste	3
Justification of choice of data or description of measurement methods and procedures applied	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Table 2.4)	
Purpose of Data	For the calculation of the project emission	
Comments	NA	

Data / Parameter	Dry matter content in % of wet weight
Data unit	%
Description	Dry matter content in % of wet weight
Source of data	Table 2.4, chapter 2, volume 5 of IPCC 2006 guidelines[43]
Value applied	

Waste type j	Dry matter content in % of wet weight
Paper/cardboard	90
Textiles	80
Food waste	40
Wood	85
Garden and Park waste	40
Nappies	40
Rubber and Leather	84
Plastics	100
Metal	100
Glass	100
Other, inert waste	90

[43] https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/5_Volume5/V5_2_Ch2_Waste_Data.pdf

Justification of choice of data or description of measurement methods and procedures applied	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Table 2.4)
Purpose of Data	For the calculation of the project emission
Comments	NA

Data / Parameter	$NCV_{i,y}$
Data unit	TJ/Gg
Description	Weighted average net calorific value of fuel type i in year y
Source of data	Table 1.2 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories[44]
Value applied	For Diesel Oil- 43.3 TJ/Gg
Justification of choice of data or description of measurement methods and procedures applied	IPCC default values at the upper limit of the uncertainty at a 95% confidence interval as provided in Table 1.2 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories
Purpose of Data	For the calculation of the project emission
Comments	NA

Data / Parameter	$EF_{CO2,i,y}$
Data unit	kg/TJ
Description	Weighted average CO_2 emission factor of fuel type i in year y
Source of data	Table 1.4 of Chapter1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories[45]
Value applied	For Diesel Oil- 74,800 kg/TJ

[44] https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_1_Ch1_Introduction.pdf

[45] https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_1_Ch1_Introduction.pdf

Justification of choice of data or description of measurement methods and procedures applied	IPCC default values at the upper limit of the uncertainty at a 95% confidence interval as provided in table 1.4 of Chapter of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories
Purpose of Data	For the calculation of the project emission
Comments	NA

Data / Parameter	$\rho_{i,y}$
Data unit	kg/m^3
Description	Weighted average density of fuel type i in year y
Source of data	Regional or national default values
Value applied	For diesel oil- 880 kg/m^3
Justification of choice of data or description of measurement methods and procedures applied	The density of diesel has been sourced from the data published by Indian Oil.[46]
Purpose of Data	For the calculation of the project emission
Comments	NA

Data / Parameter	$EF_{CO2,f}$
Data unit	g CO_2/t km
Description	Default CO_2 emission factor for freight transportation activity f
Source of data	Default
Value applied	<table><tr><th>Vehicle class</th><th>Emission factor (g CO_2/t km)</th></tr><tr><td>Light vehicles</td><td>245</td></tr></table>
Justification of choice of data or description of measurement methods and procedures applied	The vehicles used for freight transportation in the project has a Gross Vehicle Mass less than 26 tonnes which comes under light vehicles category. Hence, CO_2 emission factor for freight transportation activity has been considered as 245 g CO_2/t km.

[46] https://iocl.com/light-diesel-oil

Purpose of Data	For the calculation of the project emission
Comments	NA

5.2 Data and Parameters Monitored

Data / Parameter	$EG_{PJ,y}$
Data unit	MWh/Year
Description	Quantity of net electricity generation supplied by the project (Waste to Energy) plant/unit to the grid in year y
Source of data	Energy generation Reports
Value applied	Net Power Generation 77,616 MWh/year
Justification of choice of data or description of measurement methods and procedures applied	Energy meters Q0833070 and Q0833075 ("Standard Summator-100") of accuracy class 0.2s The Net electricity is calculated based on Exported electricity-Imported electricity. Monthly meter readings are taken from the main and check meter installed at metering point and certified by the representatives of electricity officials and the representatives of the project proponent.
Purpose of Data	For the calculation of the baseline Emission
Comments	The details of energy meters are provided below: Serial No.: 1) Q0833070 2) Q0833075 Calibration frequency: Once in 5 years[47] Date of Calibration: 07-July-2023

Data / Parameter	Electricity consumed by the project activity
Data unit	MWh/Year
Description	Quantity of net electricity consumed by the project activity in year y
Source of data	Energy consumption Records

[47] https://cea.nic.in/wp-content/uploads/2020/02/meter_reg.pdf

Value applied	Electricity consumed (Auxiliary) – 11,088 MWh/year (12.5%)
Justification of choice of data or description of measurement methods and procedures applied	Energy meters Q0833070 and Q0833075 ("Standard Summator-100") of accuracy class 0.2s The electricity consumed for the plant operations will be measured from energy meter.
Purpose of Data	For the calculation of the baseline emission
Comments	The auxiliary electricity consumption for RDF based MSW project is calculated as 12.5% of the gross electricity generation by the project activity. The details of energy meters are provided below: Serial No.: 1) Q0833070 2) Q0833075 Calibration frequency: Once in 5 years[48] Date of Calibration: 07-July-2023

Data / Parameter	W_x
Data unit	tonnes per day
Description	Total amount of waste disposed in a SWDS in year x or month i
Source of data	Weigh Bridge & Load cell Records
Value applied	700
Justification of choice of data or description of measurement methods and procedures applied	The MSW load transported to the project site is measured using weigh bridge which will be calibrated annually as per the monitoring plan laid out by the project proponent. The quantity of waste delivered will be monitored on daily basis and recorded. The waste quantity of around 1200 TPD received at the plant will be preprocessed, out of which 700 TPD is utilized for the plant operation and also be considered as total amount of waste disposed in a SWDS for which baseline emissions has been calculated. Calculation method: Net weight of MSW Load = Gross weight of the truck with MSW load - Tare weight of empty truck

[48] https://cea.nic.in/wp-content/uploads/2020/02/meter_reg.pdf

	W_x (Applicable combustible properties, 700 TPD) = Net weight of MSW Load – Inert waste – Waste stream for compost plant
Purpose of Data	For the calculation of the baseline emission and project emission
Comments	The weigh bridge calibration details are provided below:

Serial No.	1) R0838/1 2) R0839/1
Calibration frequency	Once in 12 months
Date of Calibration/ validity	04-February-2023/03-February-2024 31-January-2024/ 30- January -2025
Certificate No.	CLM17722469

Data / Parameter	$p_{n,j,x}$
Data unit	%
Description	Weight fraction of the waste type j in the sample n collected during the year x
Source of data	Sampling Records
Value applied	(see table below)
Justification of choice of data or description of measurement methods and procedures applied	Random samples are collected after the pre-processing section (i.e.) from the MRF, out of which representative sample of more than 25 kg is taken for analysis. The waste composition based on the waste types j are segregated and weighed for each waste fraction (measured on wet basis).

Description	% composition of MSW
Food	0.00%
Paper	19.71%
Wood including Coconut Shell	2.20%
Green/Mix	46.39%
Plastic	8.70%
Cloth	20.59%
Rubber	0%
Metal	0%
Glass	0.70%
Inert	1.71%
Total	**100%**

	Sampling is done on quarterly basis with minimum of three samples.
Purpose of Data	For the calculation of the baseline emission and project emission
Comments	NA

Data / Parameter	z_x
Data unit	-
Description	Number of samples collected during the year x
Source of data	Sampling Records
Value applied	Minimum of three samples every three months
Justification of choice of data or description of measurement methods and procedures applied	The pre-processed MSW (RDF) will be sampled minimum of three samples every three months.
Purpose of Data	For the calculation of the baseline emission and project emission
Comments	NA

Data / Parameter	$Q_{,y}$
Data unit	Tonnes per day (TPD)
Description	Amount of waste combusted, gasified or mechanically/thermally treated by the project activity in each year y
Source of data	Load cell records
Value applied	700 TPD
Justification of choice of data or description of measurement methods and procedures applied	The quantity of RDF fed to the furnace has been measured through the load cells installed in the grab crane. The quantity of RDF consumed has been monitored on daily basis and recorded. The load cell in the grab crane has been subjected to periodic calibration (in accordance with the manufacturers specification).
Purpose of Data	For the calculation of project emission
Comments	NA

Data / Parameter	$Q_{,y,ash}$
Data unit	Tonnes per day (TPD)
Description	Quantity of combustion residues generated by the project activity in each year y
Source of data	Plant Records
Value applied	77.38
Justification of choice of data or description of measurement methods and procedures applied	The residual waste from the controlled combustion process includes fly ash and bottom ash. The bottom ash and fly ash have been sent to Construction & Demolition waste processing plant. The quantity of fly ash and bottom ash is measured through the weighing machine. The quantity of residues generated from the combustion process will be monitored on daily basis and recorded.
Purpose of Data	For the calculation of project emission
Comments	The weigh bridge calibration details are provided below: <table><tr><td>Serial No.</td><td>1) R0838/1 2) R0839/1</td></tr><tr><td>Calibration frequency</td><td>Once in 12 months</td></tr><tr><td>Date of Calibration/ validity</td><td>04-February-2023/03-February-2024 31-January-2024/ 30- January -2025</td></tr><tr><td>Certificate No.</td><td>CLM17722469</td></tr></table>

Data / Parameter	$Q_{y,non-biomass}$
Data unit	Tonnes per year
Description	Quantity of non-biomass carbon content of the RDF combusted in each year y
Source of data	Sampling Records
Value applied	19,173 tonnes per year
Justification of choice of data or description of measurement methods and procedures applied	Random samples are collected after the pre-processing section (i.e.) MRF, out of which representative sample of more than 25 kg is taken for analysis. The waste composition based on the waste types j are segregated and weighed for each waste fraction. (measured on wet basis). The quantity of non-biomass carbon

	content is calculated based on the IPCC default values of total carbon content (FCC) and fossil carbon content (FFC). Sampling is done on quarterly basis with minimum of three samples.
Purpose of Data	For the calculation of project emission
Comments	NA

<table>
<tr><td>Data / Parameter</td><td>CT_y</td></tr>
<tr><td>Data unit</td><td>Tonnes/truck</td></tr>
<tr><td>Description</td><td>Average truck capacity for transportation</td></tr>
<tr><td>Source of data</td><td>Weigh Bridge Records</td></tr>
<tr><td>Value applied</td><td>--</td></tr>
<tr><td>Justification of choice of data or description of measurement methods and procedures applied</td><td>Values are monitored via the recordings of the net capacity of the trucks at weighbridge readings.</td></tr>
<tr><td>Purpose of Data</td><td>For the calculation of project emission</td></tr>
<tr><td>Comments</td><td>The weigh bridge calibration details are provided below:
<table>
<tr><td>Serial No.</td><td>1) R0838/1
2) R0839/1</td></tr>
<tr><td>Calibration frequency</td><td>Once in 12 months</td></tr>
<tr><td>Date of Calibration/ validity</td><td>04-February-2023/03-February-2024
31-January-2024/ 30-January -2025</td></tr>
<tr><td>Certificate No.</td><td>CLM17722469</td></tr>
</table>
.</td></tr>
</table>

Data / Parameter	$FC_{i,j,y}$
Data unit	Tonnes per year
Description	Quantity of fuel type i combusted in process j during the year y
Source of data	Plant Records

Value applied	3.6
Justification of choice of data or description of measurement methods and procedures applied	The project involves consumption of diesel for the waste handling vehicle, loader, DG set etc. The quantity of fuel consumed will be monitored using standard gauge daily and recorded. The data recorded can be cross checked with the fuel purchase bills.
Purpose of Data	For the calculation of project emission
Comments	NA

Data / Parameter	$Q_{y,fuel}$
Data unit	tonnes
Description	Quantity of auxiliary fossil fuel used in the year y
Source of data	Measured
Value applied	25.72
Justification of choice of data or description of measurement methods and procedures applied	The quantity of fuel consumed for combustion of RDF in furnace will be monitored on daily basis and recorded.
Purpose of Data	For the calculation of project emission
Comments	NA

Data / Parameter	Distance for transporting the waste in the baseline and the project scenario
Data unit	km
Description	The distance of municipal solid waste transportation in the baseline scenario and in the project scenario.
Source of data	Plant Records
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	The average distance travelled by the truck is calculated through the odometer readings.

Purpose of Data	For the calculation of project emission
Comments	The collection and transportation of MSW is under the scope of Pimpri- Chinchwad Municipal Corporation. The average distance between the project site and waste collection point is in same range of distance between the waste collection point and the landfill site in the baseline scenario. Hence the project emissions from freight transportation for collection of MSW is considered as zero.

Data / Parameter	$D_{f,m}$
Data unit	km
Description	Return trip distance between the origin and destination of freight transportation activity f in monitoring period m
Source of data	Plant Records
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	The average distance travelled by the truck is calculated through the odometer readings.
Purpose of Data	For the calculation of project emission
Comments	The collection and transportation of MSW is under the scope of Pimpri- Chinchwad Municipal Corporation. The average distance between the project site and waste collection point is in same range of distance between the waste collection point and the landfill site in the baseline scenario. The fly ash and bottom ash has been sent to Construction & Demolition processing plant which is in the vicinity of the project site and the inert waste has been disposed in the Landfill which is in project site. Hence the project emissions from freight transportation for collection of MSW, inert waste and residual ash is considered as zero.

Data / Parameter	$FR_{f,\ m}$
Data unit	Tonnes per day
Description	Total mass of freight transported in freight transportation activity f in monitoring period m

Source of data	Weigh Bridge Records
Value applied	$FR_{MSW,m}$ – 700 $FR_{inert,m}$ – 142.85 $F_{ash,m}$ – 77.38
Justification of choice of data or description of measurement methods and procedures applied	Values are monitored via the recordings of the net capacity of the trucks at weighbridge readings and cross-checked with the vehicle licenses.
Purpose of Data	For the calculation of project emission
Comments	The weigh bridge calibration details are provided below: <table><tr><td>Serial No.</td><td>1) R0838/1 2) R0839/1</td></tr><tr><td>Calibration frequency</td><td>Once in 12 months</td></tr><tr><td>Date of Calibration/ validity</td><td>04-February-2023/03-February-2024</td></tr><tr><td>Certificate No.</td><td>CLM17722469</td></tr></table>

5.3 Monitoring Plan

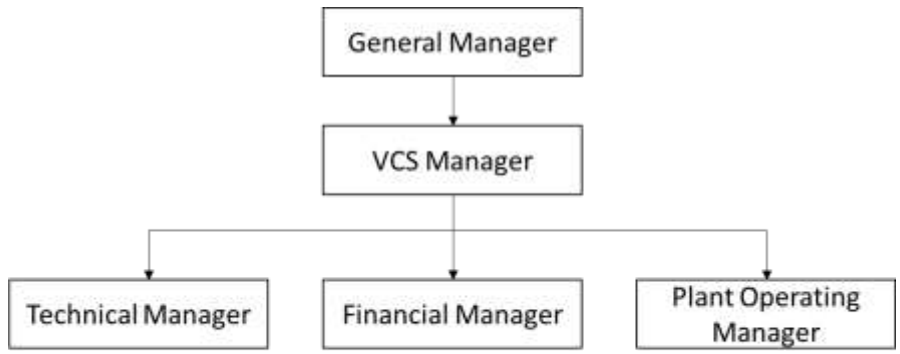
The monitoring plan is developed to establish suitable data collection method for measurement & collection of data and maintenance of records according to the monitoring methodology of AMS IIIE "Avoidance of methane production from decay of biomass through controlled combustion, gasification or mechanical/thermal treatment" Version 17.0 and AMS.I.D." Grid connected renewable electricity generation" Version 18.0. The monitoring plan is project specific for which, the project performance with all relevant criteria will be monitored. Proper training will be provided to concerned personnel for operation purpose.

The purpose of monitoring is to calculate and monitor GHG emission reduction by the project activity. The monitoring plan, which is implemented by the project proponent describes about the following aspects:

- Overall project management
- Monitoring Plan
- Emergency Preparedness
- Training on Monitoring & Archiving of Data and Internal Audit Procedures
- Quality Assurance and Quality Control (QA/QC) procedure

Overall Project Management and Team

The project proponent organizes a separate team to be responsible for data collection, supervision and witness the whole process of data measuring and recording. A senior manager will be appointed to take full responsibility for the overall monitoring of the project. The monitoring and measurement will be carried out by designated monitoring officers. The site in-charge will be responsible for carrying out internal auditing and QA/QC. All the values from generation record will be checked with the invoices for consistency.



Monitoring Plan

The monitoring procedures will be carried out as mentioned in section 5.2. The data will be recorded on a continuous basis or as indicated in section 5.2 and backup of the same will be maintained. The data to be monitored include MSW composition and quantity, electricity exported and imported, auxiliary fuel consumption, quantity of combustion residues:

S. No	Parameter
1	MSW composition ($P_{n,j,x}$) will be analyzed in accordance with the required standards. Random samples are collected after the pre-processing section (i.e.) from the MRF, out of which representative sample of more than 25 kg is taken for analysis. The waste composition based on the waste types j are segregated and weighed for each waste fraction (measured on wet basis). Sampling is done on quarterly basis with minimum of three samples.
2	The amount of waste (Q_y) to be combusted will be measured and recorded continuously through the load cells installed at the grab crane. The quantity of RDF consumed has been monitored on daily basis and aggregated monthly. The load cells in the grab crane have been subjected to periodic calibration (in accordance with the manufacturers specification).
3	The amount of combustion ($Q_{y,ash}$) residues will be measured and recorded continuously through weigh bridge records based on the number of truck loads dispatched from the plant to the C&D plant. The data will be aggregated monthly and annually.
3	Electricity ($EG_{pj,y}$) delivered to power grid by the project will be recorded by the electricity meter. Copies of receipts or invoices will be provided to the VVB for validation.

4	The grid electricity consumed by the project will be monitored using the electricity meter. Copies of receipts or invoices will be provided to the VVB for validation.
5	The quantity of auxiliary fossil fuel consumed in the project activity will be monitored daily and recorded. Copies of fuel purchase receipts or invoices will be provided to the VVB for validation.
6	The quantity of non-biomass carbon content ($Q_{y,non-biomass}$) of the RDF combusted will be measured through sampling. Sampling is done on quarterly basis with minimum of three samples. Random samples are collected after the pre-processing section (i.e.) from the vibro feeder, out of which representative sample of more than 25 kg is taken for analysis. The waste composition based on the waste types j are segregated and weighed for each waste fraction. (measured on wet basis). The quantity of non-biomass carbon content of the RDF is calculated based on the IPCC default values of total carbon content (FCC) and fossil carbon content (FFC).
7	The distance for transportation of waste and the average truck capacity will be monitored and recorded.
8	Number of samples (Z_x) collected during the year x will be monitored to analyse the RDF composition. Sampling is done on quarterly basis with minimum of three samples.

In addition, the SDG parameters stated in section 1.17 will be monitored.

Personal Training:

The project employs qualified and experienced persons for plant operation. The training period is crucial for ensuring that employees are knowledgeable about the plant's operations, safety protocols, environmental regulations, and efficient use of resources. The project proponent prioritizes safety training to ensure that all employees understand and follow safety protocols and procedures, including emergency response plans. This includes training on handling hazardous materials, operating equipment safely, and preventing accidents. The training course will be thoroughly and meticulously designed, highlighting the objectives, salient features, operational aspects and trouble shooting.

Emergency preparedness:

In case of any unforeseen event that is not covered under this monitoring plan, staff of the operation division will immediately inform the chief of the operation division. The chief of the operation division is then responsible to ensure that the cause for the unforeseen event is detected, the event is remedied and for the period of time in which the unforeseen event has occurred uncertainty in data gathered is limited as much as possible.

Internal auditing

The internal auditing process involves a systematic and objective evaluation of the plant's operations, processes, and controls to ensure compliance with regulations, identify operational efficiencies, mitigate risks, and improve overall performance. The project proponent will have a team of internal expertise in waste-to-energy technology, environmental regulations, health and safety, and financial management. Project proponent will conduct the internal auditing by cross checking the data from the invoices and data logbooks for ensuring the quality and consistency of data measured and monitored for the purpose of emission reduction calculations.

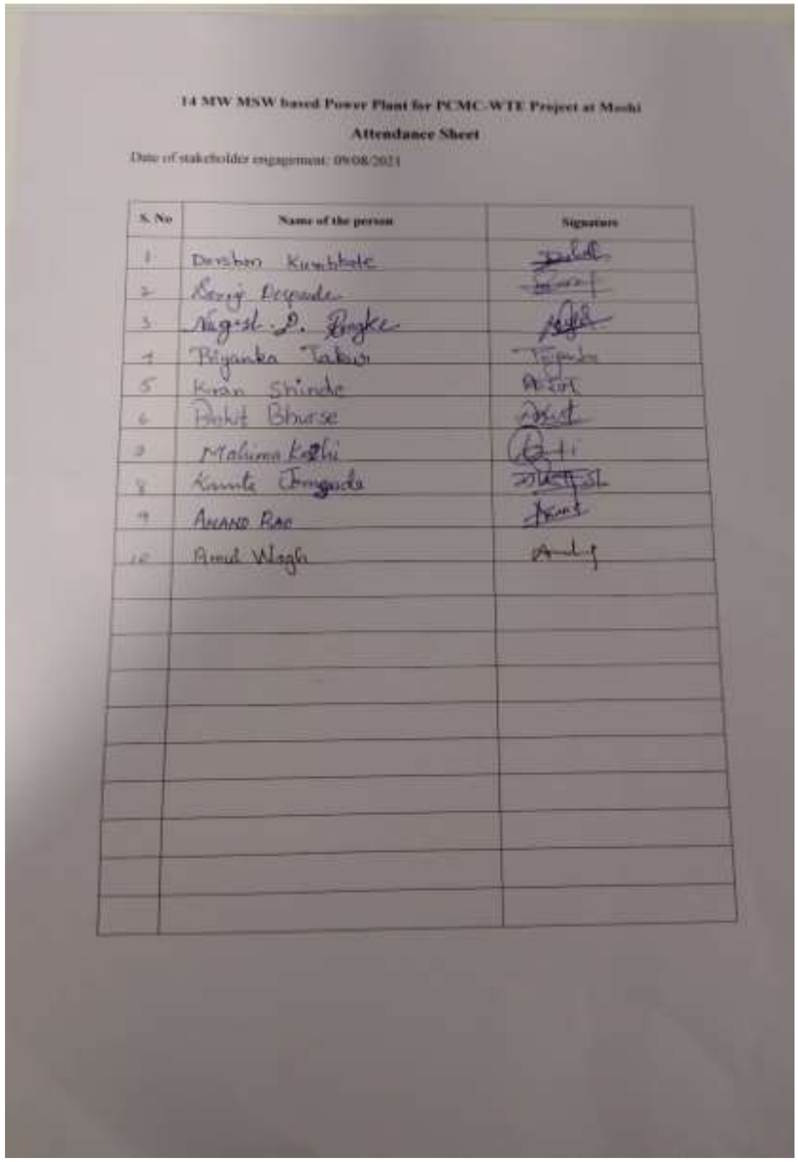
Quality Assurance and Quality Control (QA/QC) procedures
Calibration of Measuring Equipment's:

The reliability of the monitoring system depends on the accuracy of the measuring instrument and the quality of the relevant equipment. Thus, all the meters will be regularly calibrated as per the frequency specified by the manufacturer. To assist in future verifications, the project proponent will preserve the calibration records, along with the data files of project monitoring. Error check routines will be established on site and at the point of data storage to detect data measuring / transmission failures as well as malfunctions. In the case of malfunction of the meters, the meter supplier will provide technical support to engage the problem promptly and emission reductions during the corresponding period will be calculated conservatively. The installation of the electricity metering equipment will fulfil the requirements of the relevant national standard.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Not Applicable

APPENDIX 2: DETAILS OF LOCAL STAKEHOLDER CONSULTATION HELD



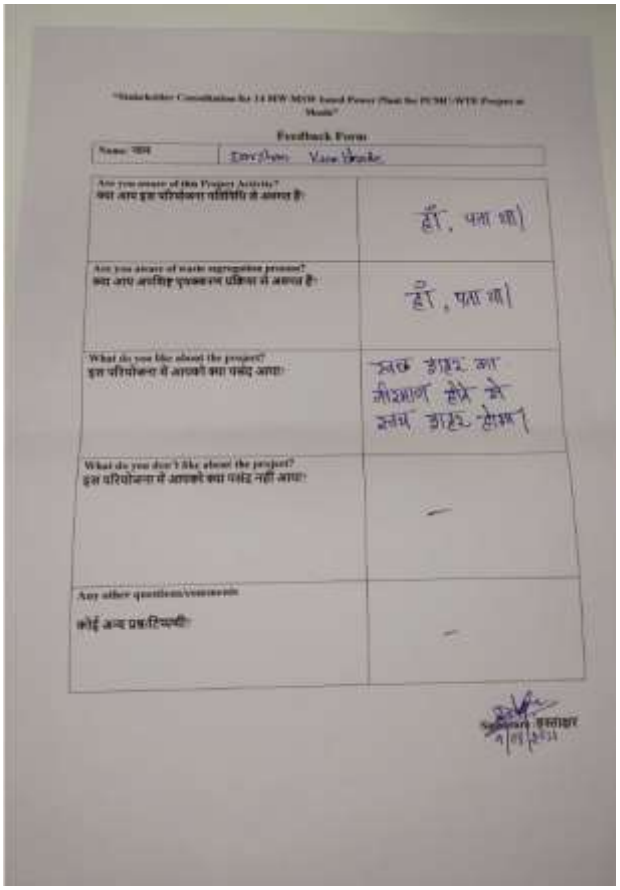
14 MW MSW based Power Plant for PCMC-WTE Project at Moshi

Attendance Sheet

Date of stakeholder engagement: 09/08/2021

S. No	Name of the person	Signature
1	Darshan Kumbhate	[illegible]
2	Saroj Despande	[illegible]
3	Nagesh D. Bagke	[illegible]
4	Priyanka Takur	[illegible]
5	Kiran Shinde	[illegible]
6	Rohit Bhurse	[illegible]
7	Mahima Kolhi	[illegible]
8	Kavita Jangade	[illegible]
9	ANAND RAO	[illegible]
10	Amol Wagh	[illegible]

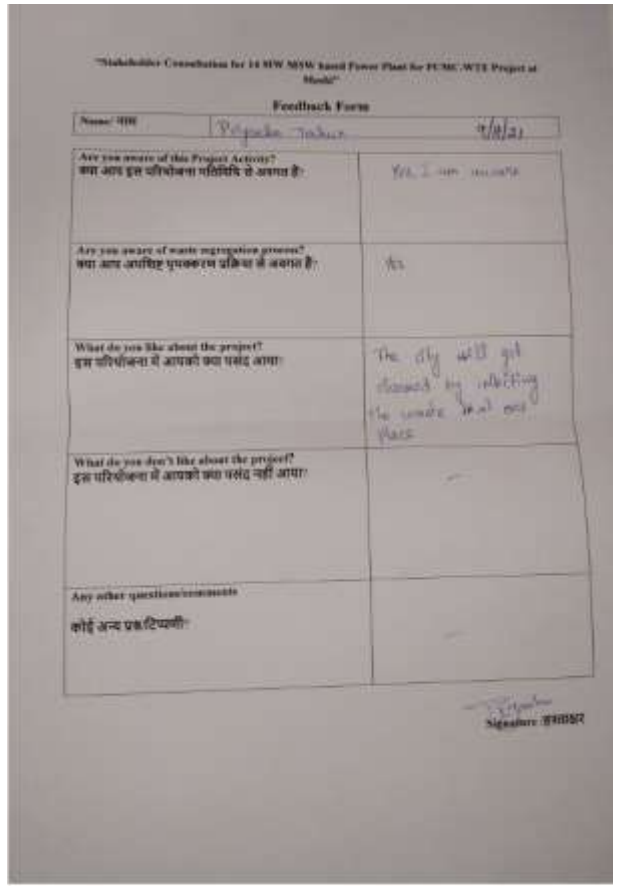
Figure 1: Attendance sheet of Local Stakeholders Consultation Meeting 1


"Stakeholder Consultation for 14 MW MSW based Power Plant for PCMC WTE Project at Moshi"

Feedback Form

Name/ नाम	
Are you aware of this Project Activity? क्या आप इस परियोजना गतिविधि से अवगत है	हाँ, पता था।
Are you aware of waste segregation process? क्या आप अपशिष्ट पृथक्करण प्रक्रिया से अवगत है	हाँ, पता था।
What do you like about the project? इस परियोजना में आपको क्या पसंद आया	
What do you don't like about the project? इस परियोजना में आपको क्या पसंद नहीं आया	-
Any other questions/comments कोई अन्य प्रश्न/टिप्पणी	-

Signature हस्ताक्षर


"Stakeholder Consultation for 14 MW MSW based Power Plant for PCMC WTE Project at Moshi"

Feedback Form

Name/ नाम	
Are you aware of this Project Activity? क्या आप इस परियोजना गतिविधि से अवगत है	Yes, I am aware
Are you aware of waste segregation process? क्या आप अपशिष्ट पृथक्करण प्रक्रिया से अवगत है	Yes
What do you like about the project? इस परियोजना में आपको क्या पसंद आया	The city will get cleaned by collecting the waste at one place
What do you don't like about the project? इस परियोजना में आपको क्या पसंद नहीं आया	-
Any other questions/comments कोई अन्य प्रश्न/टिप्पणी	-

Signature हस्ताक्षर

Figure 2: Feedback forms of Local Stakeholders Consultation Meeting 1

"Local Stakeholder Consultation Meeting for
"14 MW Waste-to-Energy Project of Antony Lara Renewable Energy Private Limited
awarded by Pimpri Chinchwad Municipal Corporation"
Attendance Sheet

Project Proponent : Antony Lara Renewable Energy Private Limited

Date of meeting : November 3rd, 2023

S. No.	Name of the person	Signature
1	Sanny Mane	S.Amane
2	Ashok Sudhare	Ashok
3	Santosh Tolmane	[illegible]
4	Shubham Narawade	Shubham
5	Abhay Gaikwad	[illegible]
6	Vishal Ture	[illegible]
7	वामन कावळे	वामन कावळे
8	Vikas Gaikwad	[illegible]
9	Kailas Gaikwad	कैलास
10	Nikhil Gaikwad	Nikhil
11	Swapnil Gaikwad	[illegible]
12	Rohit Akulwar	[illegible]
13	Amol Gadhak	[illegible]
14	Rohan Sanjay Hajare	[illegible]
15	Sanjay Hajare	[illegible]
16	मधुकर बोराटे	[illegible]
17	गुड्डी	गुड्डी
18	गोरा	गोरा
19	मनिषा	मनिषा
20	सोनू	सोनू
21.	कमला	कमला

Figure 3: Attendance sheet of Local Stakeholders Consultation Meeting 2


Figure 4: Local Stakeholders Consultation Meeting 2

ANTONY LARA RENEWABLE ENERGY PRIVATE LIMITED

ALR

ALRE/PCMC/WTE/0458/2022-23

Date: 25-10-2023

To,
The Honourable Municipal Commissioner,
Commissioner Office,
Pimpri Chinchwad Municipal Corporation,
Pimpri-18.

Subject: Invitation for Local Stakeholder Consultation (LSC) meeting of GHG mitigation project "14 MW Waste-to-Energy Project of Antony Lara Renewable Energy Private Limited awarded by Pimpri Chinchwad Municipal Corporation

Dear Sir,
This is regarding 14MW Waste to Energy Project at Moshi Kachara Depot. The Project has achieved its COD2 on 7th October 2023 and is commercially exporting power to Authority via state grid. The power generated is consumed at the Ravet Water Treatment plant and Charoli Sewage Treatment plant.

Presently, we are undertaking registration of the project under Verified Carbon Standard mechanism. In this regard, we are conducting a local stakeholders consultation meeting (public meeting) at Venue: Moshi Kachara Depot, Pimpri, Pune, India on November 3rd , 2023 at 11:00 A.M to take their views/suggestions on the project's social and environmental impacts.

We request your presence for this deliberation.

Yours Sincerely,
For Antony Lara Renewable Energy Pvt Ltd

Authorised Signatory

Date: October 31, 2023

To,

Honourable Municipal Commissioner,
Pimpri Chinchwad Municipal Corporation

Sub: Invitation for Local Stakeholder Consultation (LSC) meeting of GHG mitigation project "14 MW Waste-to-Energy Project of Antony Lara Renewable Energy Private Limited awarded by Pimpri Chinchwad Municipal Corporation"

Antony Lara Renewable Energy Private Limited has established Waste to Energy Power Plant at Moshi Kachara Depot, Pimpri, Pune, India awarded by Pimpri Chinchwad Municipal Corporation on DBOOT basis. The plant utilizes Refuse-Derived Fuel ("RDF") as feedstock derived from the separation and sorting of Municipal Solid Waste ("MSW") excluding the recyclable or potentially hazardous materials from the waste stream. The MSW will be sourced from the Pimpri Chinchwad Municipality area as per concession agreement with Pimpri Chinchwad Municipal Corporation. The main purpose of this project activity is to generate clean form of electricity through waste to energy-based power generation. The project will treat about 700 ton of RDF per day and generate electricity with the installed capacity of 14 MW.

The project activity will reduce the associated GHG emission reductions generated from the use of grid connected power plants in the absence of the project scenario. In addition, the project will mitigate the methane emitted into the atmosphere from the landfill, where the waste is left to decay in anaerobic conditions.

Antony Lara Renewable Energy Private Limited is undertaking registration of the project under Verified Carbon Standard mechanism. In this regard Antony Lara Renewable Energy Private Limited is conducting a local stakeholders consultation meeting (public meeting) at Venue: Moshi Kachara Depot, Pimpri, Pune, India on November 03rd, 2023 at 11.00 A.M to take their views/suggestions on the project's social and environmental impacts.

We extend you an invitation to attend the stakeholder meeting and offer your views and valuable feedback on the project's social environmental and sustainable impacts.

Kindly contact us for any further information on the subject at the address given below.

It is also informed that whoever are unable to attend the meeting in physical, are requested to provide their feedback by contacting through email or phone number provided below.

Accessibility & Contact Details :

Nearest Airport : Pune (PNQ) Airport
Nearest Railway Station : (CCH)CHINCHVAD and (PMP)PIMPRI

Our Contact Details (at Venue):
Name: Brijesh Kumar, Project Head
Organization: Antony Lara Renewable Energy Private Limited
Phone: +91- 9137526281
E-mail: brijesh.kumar@antonywaste.in

Best Regards,
Antony Lara Renewable Energy Private Limited

Figure 5: Invitation Letter to Commissioner Office

ANTONY LARA RENEWABLE ENERGY PRIVATE LIMITED

CIN : [illegible]

ALR

ALRE/PCMC/WTE/0458/2023-23

Date: 31-10-2023

To,
Sub-Regional Officer,
Maharashtra Pollution Control Board,
Jog Centre, Building, 2nd floor, Wakdewadi,
Old Mumbai-Pune Highway,
Pune 411003

Subject: Invitation for Local Stakeholder Consultation (LSC) meeting of GHG mitigation project "14 MW Waste-to-Energy Project of Antony Lara Renewable Energy Private Limited awarded by Pimpri Chinchwad Municipal Corporation

Dear Sir,
This is regarding 14MW Waste to Energy Project at Moshi Kachara Depot. The Project has achieved its COD2 on 7th October 2023 and is commercially exporting power to Authority via state grid. The power generated is consumed at the Ravet Water Treatment plant and Chikhli Sewage Treatment plant.

Presently, we are undertaking registration of the project under Verified Carbon Standard mechanism. In this regard, we are conducting a local stakeholders consultation meeting (public meeting) at Venue: Moshi Kachara Depot, Pimpri, Pune, India on November 3rd, 2023 at 11.00 A.M to take their views/suggestions on the project's social and environmental impacts.

We request your presence for this deliberation.

Yours Sincerely,
For Antony Lara Renewable Energy Pvt Ltd

Authorised Signatory

Clerk
R. O. M. P. C. B. Pune

Reg. Off. : [illegible]

Date: October 31, 2023

To,
Sub-Regional Officer,
Maharashtra Pollution Control Board.

Sub: Invitation for Local Stakeholder Consultation (LSC) meeting of GHG mitigation project "14 MW Waste-to-Energy Project of Antony Lara Renewable Energy Private Limited awarded by Pimpri Chinchwad Municipal Corporation"

Antony Lara Renewable Energy Private Limited has established Waste to Energy Power Plant at Moshi Kachara Depot, Pimpri, Pune, India awarded by Pimpri Chinchwad Municipal Corporation on DBOOT basis. The plant utilizes Refuse-Derived Fuel ("RDF") as feedstock derived from the separation and sorting of Municipal Solid Waste ("MSW") excluding the recyclable or potentially hazardous materials from the waste stream. The MSW will be sourced from the Pimpri Chinchwad Municipality area as per concession agreement with Pimpri Chinchwad Municipal Corporation. The main purpose of this project activity is to generate clean form of electricity through waste to energy-based power generation. The project will treat about 700 ton of RDF per day and generate electricity with the installed capacity of 14 MW.

The project activity will reduce the associated GHG emission reductions generated from the use of grid connected power plants in the absence of the project scenario. In addition, the project will mitigate the methane emitted into the atmosphere from the landfill, where the waste is left to decay in anaerobic conditions.

Antony Lara Renewable Energy Private Limited is undertaking registration of the project under Verified Carbon Standard mechanism. In this regard Antony Lara Renewable Energy Private Limited is conducting a local stakeholders consultation meeting (public meeting) at Venue: Moshi Kachara Depot, Pimpri, Pune, India on November 03rd, 2023 at 11.00 A.M to take their views/suggestions on the project's social and environmental impacts.

We extend you an invitation to attend the stakeholder meeting and offer your views and valuable feedback on the project's social environmental and sustainable impacts.

Kindly contact us for any further information on the subject at the address given below.

It is also informed that whoever are unable to attend the meeting in physical, are requested to provide their feedback by contacting through email or phone number provided below.

Accessibility & Contact Details -

Nearest Airport	: Pune (PNQ) Airport
Nearest Railway Station	: (CCH)CHINCHVAD and (PMP)PIMPRI

Our Contact Details (at Venue):
Name: Brijesh Kumar, Project Head
Organisation: Antony Lara Renewable Energy Private Limited
Phone: +91- 9137526281
E-mail: brijesh.kumar@antonywaste.in

Best Regards,
Antony Lara Renewable Energy Private Limited

Figure 6: Invitation Letter to MPCB